Daily Growth at Risk: financial or real drivers?

The answer is not always the same

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Abstract

We propose daily Growth-at-Risk (GaR) for monitoring downside risks of the US economy, which use financial and real high frequency indicators. We show that the relative importance, in terms of forecasting power, of these indicators is time varying. Indeed, the optimal forecasting weights of our variables were clearly different during the Global Financial Crisis and the recent Covid-19 crisis, which reflects the dissimilar nature of the two crises. We introduce LASSO, and elastic-net-adaptive-sparse-group-LASSO into the family of mixed data sampling models used to estimate GaR and show that these methods outperform past candidates explored in the literature. Moreover, equity market volatility (VXO), credit spreads and the Aruoba-Diebold-Scotti business conditions index (ADS) are found to be relevant to forecast economic activity, especially during crisis episodes. Overall, our results show that daily information for both real and financial variables is key for producing accurate point and tail risk nowcasts of economic activity.

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1. Introduction

There is a large body of recent studies that analyze the predicting power of financial conditions on real economic activity in times of crisis. One group emphasizes the significant role of financial indicators in forecasting low quantiles of the GDP growth rate (e.g. Giglio et al., 2016; Adrian et al., 2019), while others report that after controlling for real variables, financial indicators have little to add to the mix (e.g. Reichlin et al., 2020; Plagborg-Møller et al., 2020). Indeed, some studies claim the opposite, i.e., that real variables (except for unemployment claims) have little to add after financial variables have been incorporated into the forecasting equation (i.e. Carriero et al., 2020).

This disagreement occurs because forecasting real economic activity (or any fragment of the growth distribution for that matter) using financial variables is a unique and challenging problem due to two facts. First, financial and real variables are generally sampled at different frequencies, being the former of a considerably higher frequency than the latter. Second, quantifying how much financial variables add in terms of forecasting power for predicting economic activity, seems to be rather a causal question (in opposition to a predictive one), because it is related to the prolonged controversy in economics about the dichotomy between nominal and real variables, and how (and to what extent) the former influence the latter. Furthermore, this second issue highlights the tension that exists between pure "predictive" and pure "causal" tasks in social science, and in particular, in economics (Athey, 2017). In short, this theoretical distinction is not crystal clear when it comes to macroeconomic studies, which are observational out of necessity, and in which forecasting can generally be improved by using domain knowledge, which is causal. Moreover, at the same time, forecasting exercises are generally expected to improve our understanding about the causal mechanisms in the economy. In short, we trust forecasts more than we can understand.

Due to this complex relationship and the multiplicity of aims that a researcher or policy maker may have for making a forecast, we recommend using an eclectic approach. In this approach both financial and real variables should ideally be used to forecast economic crisis episodes, while letting the data show freely the relative importance of each set of variables on a time-varying basis. Following this approach we make two important contributions to the field. First, we show that the informational content of financial and real economy indicators is different across time. Thus, in some circumstances, forecasting accuracy is due greatly to financial

indicators, such as the equity market volatility index (VXO); however, in other circumstances, real economic indicators, such as the Aruoba-Diebold-Scotti business conditions index (ADS) daily business cycle index (Aruoba et al., 2009), are more useful for improving the forecast. Our results clearly show the time-varying importance of real and financial variables. We estimate the optimal weight that our forecasting models assign to ADS or to finance variables, and we show that in periods like the aftermath of the Global Financial Crisis (GFC), financial indicators play a far greater role than ADS, while the opposite it true for the recent Covid-19 crisis. This is in agreement with the general consensus in the macro-financial literature that highlights the financial nature of the GFC, when financial markets and intermediaries acted as amplifiers of shocks to the system (Isohätälä et al., 2016; Brunnermeier and Sannikov, 2016; Gertler and Gilchrist, 2018). However, past literature recognizes that the Covid-19 crisis was simply a product of the supply restrictions imposed to contain the pandemic, which were real and supply-side in nature, albeit with repercussions for the aggregate demand (Guerrieri et al., forthcoming). Note that understanding the mechanisms underlying a crisis episode is a purely causal task, which can assist the researcher to interpret the results of the forecasting exercise and also to improve the actual forecasting.

Second, we also make a contribution regarding the abovementioned issue when real economic activity is forecast using financial variables, which are sampled at higher frequencies (e.g. daily, weekly) than GDP and other real variables (e.g. quarterly, annually). Unlike most of the literature that employs weekly indicators to forecast GDP growth (or lower-frequency financial indicators), we estimate our models using *daily* right-hand-side variables. Thus, our results use more information *in time* than most of the extant literature. There are some exceptions in which daily data is also used (i.e. Ferrara et al. 2021); however, in these cases, real variables have been neglected and the number of financial indicators included is rather limited. Therefore, our results are also supported by richer *cross-sectional* information at the intended frequency than previous studies, which is reflected in the greater accuracy of our models compared to previous ones.

An important concern needs to be addressed when working with daily predictors and without excluding variables (i.e. financial or real) a priori from our set of predictors, as we do. The number of parameters to estimate increases quickly, compared to using lower frequencies and fewer variables. In this case, shrinkage, regularization and dimensionality reduction techniques, such as those provided by LASSO, the Elastic Net method or PCA, become essential. Thus, we

introduce these methods into mixed data sampling (MIDAS)-Quantile models for estimating Growth-at-Risk. We use quantile regression for high dimensional spaces as proposed by Belloni and Chernozhukov (2011) and PCA to reduce the dimensions of our problem even further. In line with what has been emphasized by Lima et al. (2020) and Lima and Meng (2017), parameter-reduction techniques are crucial at high frequencies. LASSO-Quantile and Elastic Net-Quantile models tend to outperform other alternatives explored in past literature (i.e., the traditional quantile MIDAS, where the vector of high-frequency terms takes an arbitrary form, either estimated by frequentist (Ghysels et al., 2016) or Bayesian methods (Mogliani and Simoni, 2021; Ferrara et al.,2021)), both in and out-of-sample. In addition, in line with Stock and Watson (2004), Andreou et al. (2013), and Ferrara et al. (2021) we show that combined forecasts using all indicators are more accurate, especially out-of-sample.

We validate our conclusions using a battery of statistics that come from the different fields of forecasting and quantitative risk-management. Rather than relying on a single statistic coming from the forecasting literature which, for instance, may not take into account that when the Value at Risk of a series is estimated (i.e. a low or high conditional quantile), two properties need to be achieved: Unconditional coverage and independence (Christoffersen, 1998). This is important because, as shown by Brownless and Souza (2021) in a multicountry setting, original indicators of Growth at Risk frequently do not pass basic tests designed in finance to measure the precision of VaR estimates. This, in turn, may call into question the usefulness of the entire enterprise.

We show that this is not the case for our indicators. In fact, on the vast majority of occasions, daily financial information together with daily information on real activity are very useful for anticipating bad scenarios for GDP growth. In addition, our GaR statistics are adequate and meet the performance expected from them.

The rest of this document is organized as follows: Section 2 presents our data and section 3 our methodology. Section 4 presents our main results, while section 5 concludes.

2. Data

In our exercise, we are interested in nowcasting the conditional tail of the distribution of real GDP growth (or GNP for some sample period[1]) on a pseudo-real-time basis. To do this, we use the quarterly real-time data set collected by the Federal Reserve Bank of Philadelphia (FRBP) from 1986Q1 to 2020Q4. Specifically, this dataset gathers the advance estimate for the previous quarter, which is released near the end of the month of the current quarter[2].

Regarding the daily high-frequency variables, we include 12 daily predictors to evaluate the GaR forecast (11 financial variables and 1 real variable). From this set, eight series are the same as in Pettenuzzo et al. (2016): i) the ADS daily business cycle variable of Aruoba et al. (2009), ii) the interest rate spread between the 10-year government bond rate and the federal fund rate (ISPREAD), iii) the change in the effective federal funds rate (EEFR), iv) the BAA-AAA corporate bond yield credit spread (CSPREAD), v) the excess return on the market (RET), vi) the returns on the portfolio of small minus big stocks (SMB), vii) the returns on the portfolio of high minus low book-to-market ratio stocks (HML), and viii) the returns on a winner minus loser momentum spread portfolio (MOM). In addition to this group, we add four financial indicators: the equity market volatility index VXO, which has been used as a price of risk indicator[3]; the spread between the yield of 10-year constant maturity Treasury bonds and of 3-month Treasury bills (TERM), as a predictor of US recessions[4], the spread between 3-month Libor based on US dollars and 3-Month Treasury Bill spread (TED), as a proxy of credit risk[5]; and the Composite Indicators of Systemic Risk (CISS) for the US, which is a systemic risk measure based on 15 raw market indicators, following a computation analogous to the CISS for the Euro Area (Holló et al., 2012). The latter variable is used as a benchmark indicator in our models, as the standard GaR framework considers a composite financial condition index (Adrian et al., 2019; Figueres and Jarocinski, 2020). Our data sample spans the period from 1986Q1 to

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[1] In December 1991, the Bureau of Economic Analysis switched from reporting GNP to reporting GDP as the output measure. Also, in January 1996, they switched from reporting fixed-weight aggregation methods to chain-weight methods in GDP measure.

[2] Faust et al. (2013) use this dataset to forecast real-time measures of economic activity using Bayesian Model Averaging with a large number of real and financial indicators.

[3] Rey (2015) shows that this indicator comoves with global capital flows, global credit growth, and global asset prices. Longstaff *et al.* (2011) also document that the price of sovereign risk is strongly correlated with VXO.

[4] Estrella and Hardouvelis (1998) and subsequent literature has shown the forecasting power of the term spread for recessions.

[5] Gunay (2020) show that the TED spread is superior to credit default swap indexes as an early warning indicator for the credit market.

2020Q4 and is restricted by the availability of data for all indicators. For the ADS index, we use weakly vintages starting from November 30, 2008 from the FRBP to conduct the nowcasting exercise. We include one year of daily lags of the high frequency indicator in all our specifications. The detailed description of these indicators is shown in Table 1.

Table 1. Variables

Variable	Frequency	Sample	Lags	Description	Source
ADS daily business cycle index (ADS)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	ADS index weakly vintages collected in real-time from November 30, 2008.	Federal Reserve Bank of Philadelphia
Interest rate spread (ISPREAD)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	Interest rate spread between the 10-year government bond rate and the federal fund rate	Federal Reserve Bank of St. Louis
Effective Federal Funds Rate (EEFR)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	Effective Federal Funds Rate, first difference	Federal Reserve Bank of St. Louis
Credit spread (CSPREAD)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	BAA-AAA corporate bond yield credit spread	Federal Reserve Bank of St. Louis
Excess return on the market (RET)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	Excess return on the market, value-weight return of US stocks	Fama and French (1993)
Returns on the portfolio of small minus big stocks (SMB)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	The average return on the three small portfolios minus the average return on the three big portfolios	Fama and French (1993)
Returns on the portfolio of high minus low book-to-market ratio stocks (HML)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	The average return on the two value portfolios minus the average return on the two growth portfolios	Fama and French (1993)
Returns on a winner minus loser momentum spread portfolio (MOM)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	The average return on the two high prior return portfolios minus the average return on the two low prior return portfolios	Fama and French (1993)
CBOE S&P 100 Volatility Index (VXO)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	Option-based implied volatility measure of S&P100	Federal Reserve Bank of St. Louis Federal Reserve Bank of St. Louis

Term spread (TERM)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	Spread between the yield of 10-year constant maturity Treasury bonds and of 3-month Treasury bills	Federal Reserve Bank of St. Louis
TED Spread (TED)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	Spread between 3-Month LIBOR based on US dollars and 3-Month Treasury Bill	Federal Reserve Bank of St. Louis
Composite Indicator of Systemic Stress (CISS)	Daily	Jan. 1, 1986 to Dec. 31, 2020	1 year	Daily systemic risk measure based on Holló et al. (2011)	European Central Bank
Real Gross Domestic Product (GDP)	Quarterly	q1, 1986 to q4, 2020	1 quarter	Real Gross Domestic Product or Gross National Product, Percent Change from Preceding Period, Quarterly, Seasonally Adjusted	Federal Reserve Bank of Philadelphia

Source: Authors' own elaboration.

3. Methodology

To nowcast tail risks in GDP growth, we extend the formulation of Adrian et al. (2019) to account for daily high-frequency predictors. In this section we show concisely how we adapt the standard *Growth at Risk (GaR)* to incorporate daily financial and real indicators, using mixed data sampling. To this end, we compare the performance of a traditional MIDAS (with Almon lag polynomials), Bayesian MIDAS, LASSO and Elastic Net methods. We also show how we implement combination forecasts, which have been proven to provide more accurate predictions compared to the alternatives. Finally, we present the tools we use to evaluate tail risk forecasts.

3.1. Growth at Risk framework

As in the standard framework of quarterly *Growth at Risk* pioneered by the works of Giglio et al. (2016) and Adrian et al. (2019), we rely on quantile regressions (Koenker and Bassett, 1978). Specifically, we assess the combined effect of past GDP growth (y_{t-h}) and a given financial condition indicator (x_{t-h}), at quarter t and forecast horizon h, on the current output growth (y_t). At this point it is important to recall that even tough x_{t-h} is observed daily, and it is aggregated by taking its quarter average.

The baseline quantile regression is given by:

$$y_t = \beta_0(\tau) + \beta_1(\tau)y_{t-h} + \beta_2(\tau)x_{t-h} + \epsilon_t(\tau)\,, \tag{1}$$

where $\beta(\tau) = (\beta_{0i}(\tau), \beta_0(\tau), \beta_1(\tau))'$ denotes the vector of parameters corresponding to the τ-th quantile, and $\epsilon_t(\tau)$ is a random noise.

The parameters in Equation 1 are estimated by minimizing the tick loss (TL) function:

$$TL_\tau = \frac{1}{T}\sum_t^{T-h}\left[\rho_\tau\left(y_t - \widehat{Q_\tau}(y_t|y_{t-h}, x_{t-h})\right)\right], \tag{2}$$

where $\rho_\tau(\epsilon_t) = (1-\tau)I_{\{\epsilon_t<0\}}|\epsilon_t| + \tau I_{\{\epsilon_t>0\}}|\epsilon_t|$, with $I_{\{\epsilon<0\}}$ taking the value of 1 when the subscript is true and 0 otherwise. The mathematical formulation in Equation 2 leads to the solution of a linear programming optimization problem that we have not included here. Its basic structure and the counterpart algorithm solution can be found in Koenker (2005).

The predicted value from Eq. (1) is the quantile of $y_{T|T-h}$, which is conditional on the information available up to $T-h$,

$$\widehat{Q_\tau}(y_T|y_{T-h}, x_{T-h}) = \widehat{\beta_0}(\tau) + \widehat{\beta_1}(\tau)y_{T-h} + \widehat{\beta_2}(\tau)x_{T-h}, \tag{3}$$

Koenker and Bassett (1978) further prove that the predicted value $\widehat{Q_\tau}(y_T|y_{T-h}, x_{T-h})$ is a consistent linear estimator of the conditional quantile function of y_t. In this setting, we are particularly interested in the 10 percent quantile forecast as the measure of tail risk (see Figueres et al., 2020, Ferrara et al., 2021), namely $GaR\ (10\%) = \widehat{Q_{0.10}}(y_T|y_{T-h}, x_{T-h})$[6].

This last equation can be interpreted as the 10% quantile of GDP growth, which is conditional on the information set available up to $T-h$ for the predictors. On one hand, there is a vast literature that documents that financial conditions have strong predictive information for the lower quantiles of future GDP growth (see e.g., Adrian *et al.* 2019; Prasad et al., 2019; Brownlees and Souza, 2021; Figueres et al., 2020; Ferrara et al., 2021). However, on the other hand, Plagborg-Møller et al. (2020) and Reichlin (2020) state that controlling for real factors is necessary to accurately measure the real-time effect of financial indicators on real activity. We acknowledge these two results in our framework by incorporating the ADS daily business cycle variable of Aruoba et al. (2009), in addition to the financial indicators, as a high-frequency

[6] Alternatively, Adrian et al. (2019) and Carriero et al. (2020), use the 5 percent quantile forecast as the measure of tail risk. However, due to a shorter sample compared to these works, we opt to use the 10 percent quantile.

indicator of the real sector. We then follow a combination approach to produce a better point forecast and check the optimal weights of individual high frequency predictors, following previous literature in this field (see Stock and Watson, 2004; Andreou et al., 2013; Pettenuzzo et al., 2016; Ferrara et al., 2021).

3.2. Adapting the standard GaR approach to high-frequency indicators

The problem of the formulation in Eq. (1) is that the aggregation of the high-frequency indicator prevents the model from responding to daily shocks. Consequently, similar to Ferrara et al. (2021), we adapt it to account for the daily information flow of the high-frequency indicator up to the latest available observation (minus h_d days), based on the following regression:

$$y_t = \beta_0(\tau) + \beta_1(\tau) y_{t-1} + \boldsymbol{X}_{t-h_d}^{D}{}' \boldsymbol{\phi}(\tau) + \epsilon_t(\tau)\,, \tag{4}$$

where $\phi(\tau)$ is a $p * 1$ vector of daily parameters and $\boldsymbol{X}_t^D = (x_t^1, x_t^2, \dots, x_t^p)'$ is a $p * 1$ vector of the high-frequency variable available on a daily basis, with $x_t^j, j = (1,2, \dots, p)$, which is updated d times between quarter t and $t-1$. In this setup, we consider that y_t is affected by up to one year ($q = 4$ quarters) of past daily shocks and past GDP growth, giving a total number of parameters (including the constant) approximately equal to $K = q * d + 2 = 4 * 60 + 2 = 242$, assuming a five-day working week $(d = 60\ days)$; that is, $\boldsymbol{X}_t^D = (x_t^1, x_{t-\frac{1}{60}}^2, \dots, x_{t-\frac{239}{60}}^{240})'$. Notice that in this formulation, the forecast horizon is expressed in high-frequency terms, that is, $h_d = (0,1/d, 2/d, \dots, (p-1)/d)$.

Our estimation window is wider than that of Ferrara et al. (2021), as they consider a lag window of 60 days for the high-frequency indicator. This enables the model to capture up to one year of daily information. In our case, the number of parameters K is relatively larger than the total number of observations T, so we are faced with a parameter proliferation problem, which invalidates the standard estimation procedure of the quantile regression. Thus, in what follows we discuss the four alternative methods used in our results section to estimate the above regression.

3.2.1. MIDAS-Q

The mixed data sampling (MIDAS) quantile model (MIDAS-Q) offers an effective solution to incorporating high-frequency indicators into Eq. (4). This relies on a restriction of the form in which the distributed lags of the high frequency variable are included in the regression.

Specifically, we introduce the high-frequency lagged vector $\boldsymbol{X}_{t-h_d}^{D}$ in a quantile regression for the low-frequency dependent variable y_t as follows:

$$y_t = \beta_0(\tau) + \beta_1(\tau)y_{t-1} + \sum_{i=0}^{p-1} b(i;\boldsymbol{\theta_j}(\tau))L^{\frac{i}{d}}x_t^j + \epsilon_t(\tau)\,, \tag{5}$$

where $b(i;\boldsymbol{\theta_j}(\tau)) = \sum_{k=0}^{c}\theta_{k,j}i^k$ is the Almon lag polynomial weighting function that depends on the vector of parameters $\boldsymbol{\theta_j}(\tau)$, the lag order of the high-frequency variable $i = (1,2,\dots,p-1)$, and the order of the Almon lag polynomial given by c. While Ghysels et al. (2016) propose the Beta lag polynomial function for the quantile weighting function, we consider the Almon lag polynomial[7] as in other recent works (Lima et al., 2020; and Mogliani and Simoni, 2021; Ferrara et al., 2021). Under the so-called "direct method", Eq. (5) can be reparametrized as follows:

$$y_t = \beta_0(\tau) + \beta_1(\tau)y_{t-1} + \widetilde{\boldsymbol{X}}_{t-h_d}^{D}{}'\,\boldsymbol{\phi}(\tau) + \epsilon_t(\tau)\,, \tag{6}$$

where $\widetilde{\boldsymbol{X}}_t^D = Q * \boldsymbol{X}_{t-h_d}^D$ is a $(c+1)*1$ vector representing the transformed high-frequency predictor, Q is a $((c+1)*p)$ weighting matrix with the $(i-th+1)$ row element of $\mathbf{Q}$ is equal to $\left(0^i, 1^i, 2^i, \dots, (p-1)^i\right)$. Following Ferrara et al. (2021), we set $c = 3$ (third degree Almon lag) and impose two end-point zero restrictions on the slope and the value of the lag polynomial $(r = 2)$, such as $b\left(p-1;\boldsymbol{\theta_j}(\tau)\right) = 0$ and $\nabla_p b\left(p-1;\boldsymbol{\theta_j}(\tau)\right)|_{p=p-1} = 0$, as in Mogliani and Simoni (2021). This causes the weighting structure to slowly reduce to zero. Consequently, the number of parameters to be estimated reduced substantially from $(p+1)$ to $(c+1-r)$ parameters.

3.2.2. BMIDAS-Q

The Bayesian version of the MIDAS quantile (B-MIDASQ), based on the Asymmetric Laplace Density (ALD) estimation approach pioneered by Yu and Moyeed (2001), offers a convenient alternative for estimating Eq. (4). Specially, this approach is considered by Ferrara et al. (2021) to nowcast GaR for the eurozone using high-frequency financial indicators. Yu and Moyeed (2001) showed that the minimization problem of quantile regressions (see Eq. (2)) is equivalent to maximizing a likelihood using the asymmetric Laplace distribution (ALD) for the error term

ϵ_t. We use the Gibbs sampler implementation of Kozumi and Kobayashi (2011), alongside their mixture representation.

In this framework, the error term ϵ_t in Eq. (6) can be represented as a location scale mixture of normal distributions in which the mixing distribution follows an exponential distribution (for details see, Kozumi and Kobayashi, 2011).

This implies that Eq. (4) can be expressed as:

$$y_t = \beta_0(\tau) + \beta_1(\tau)y_{t-1} + \boldsymbol{X}_{t-\mathrm{h}_d}^{D}{}' \boldsymbol{\phi}(\tau) + \varphi_1(\tau)\, v_t + \varphi_2(\tau)\sqrt{\sigma v}_t u_t\,, \qquad (7)$$

where φ_1 and φ_2 are fixed parameter functions of the quantile τ, $v_t = \sigma z_t$ in which z_t is a standard exponential function, and u_t is a standard normal function. This leads to the following likelihood distribution (to simplify assume $\boldsymbol{X_t}$ contains all covariates):

$$f(y_t|\boldsymbol{X_t'}\boldsymbol{\phi}(\tau), v_t, \sigma) \propto \exp\left(-\sum_{t=1}^{T}\frac{\left(y_t-\boldsymbol{X_t'}\boldsymbol{\phi}(\tau)-\varphi_1(\tau)v_t\right)^2}{2\varphi_2(\tau)^2\sqrt{\sigma v_t}}\right)\prod_{t=1}^{T}\frac{1}{\sqrt{\sigma v}},$$

with posterior densities for $\boldsymbol{\phi}$, v and σ given by:

$$\boldsymbol{\phi}|X, v, \sigma, \tau \sim N(\tilde{\beta}, \tilde{V}\ ),$$

$$v|X, \boldsymbol{\phi}, \sigma, \tau \sim GiG\left(\frac{1}{2}, \frac{\left(y_t-\boldsymbol{X_t'}\boldsymbol{\phi}(\tau)\right)^2}{\sigma\varphi_2(\tau)^2}, \frac{2}{\sigma}+\frac{\varphi_1(\tau)^2}{\sigma\varphi_2(\tau)}\right),$$

$$\sigma|X, v, \boldsymbol{\phi}, \tau \sim \textit{Inv-Gamma (a, b)}$$

where

$$\tilde{\beta} = \tilde{V}\sum_{t=1}^{T}\frac{\left(y_t-\boldsymbol{X_t'}\boldsymbol{\phi}(\tau)-\varphi_1(\tau)v_t\right)^2}{\varphi_2(\tau)^2\sigma v_t} \text{ and } \tilde{V}^{-1} = \sum_{t=1}^{T}\frac{\boldsymbol{X_t'}\boldsymbol{X_t}}{\varphi_2(\tau)^2\sigma v_t} + \tilde{V}_0^{-1}$$

and a and b are shape and scale parameters of the inverse gamma distribution, respectively. In our framework, as in Carriero *et al.* (2020), we are interested in using the posterior mean of the coefficient vector $\boldsymbol{\phi}(\tau)$ to produce point forecasts.

3.2.3. Soft thresholding: LASSO-Q and EN-Q

One caveat of the restricted MIDAS approach presented above is the predetermined choice of the weighting function, which may lead to a lag structure for the high-frequency predictor that does not maximize forecast accuracy. Therefore, alternatively we propose estimating GaR by

using either the LASSO or elastic net penalized methods for choosing a lag structure for the high-frequency predictors (Bai and Ng, 2008; Lima et al., 2020).

Accordingly, we select the lags of the high frequency variable, based on the LASSO quantile (LASSO-Q) algorithm proposed by Belloni and Chernozhukov (2011). The model can be summarized as follows,

$$\min_{\beta,\phi} E\left[\rho_\tau(y_t - \beta_0(\tau) - \beta_1(\tau)y_{t-1} - \boldsymbol{X}^D_{t-\mathrm{h}_d}{}'\,\phi(\tau))\right] + \lambda_\tau\left[\frac{\sqrt{\tau(1-\tau)}}{T}\right]\sum_{i=1}^{p}|\phi(\tau)_i|\,, \quad (8)$$

where the optimization problem is the sum of the standard quantile minimization function (as in Equation 2) and a penalty function given by a scaled l_1-norm of the daily vector of parameters $\phi(\tau)_i$. The overall penalty is given by $\lambda_\tau[\sqrt{\tau(1-\tau)}/T]$, where T is the sample size. The optimal level of λ_τ (LASSO penalization) is calculated as in Belloni and Chernozhukov (2011). The LASSO penalty has the distinctive feature of making the coefficients of insignificant predictors exactly equal to zero, keeping only the informative predictors for the forecast.

The elastic net (EN) estimator of Zou and Hastie (2005) aims to solve two potential drawbacks of the original LASSO. First, if $> T$, LASSO can select T variables at most. Second, if there is a group of variables with high pairwise correlation coefficients, LASSO tends to select only one variable from the group and does not care which one. Both LASSO and EN shrink the estimates and perform model selection. However, while the LASSO penalty is convex, the EN penalty is strictly convex, which means that predictors must be grouped to have similar coefficients. The elastic net quantile (EN-Q) objective function is given by:

$$\min_{\beta,\phi} E\left[\rho_\tau(y_t - \beta_0(\tau) - \beta_1(\tau)y_{t-1} - \boldsymbol{X^d}_{\boldsymbol{t-\mathrm{h}_d}}{}'\boldsymbol{\phi}(\tau))\right] +$$

$$\lambda_{1,\tau}\sum_{i=1}^{p}|\phi(\tau)_i| + \lambda_{2,\tau}\sum_{i=1}^{p}\phi(\tau)_i^2\,, \quad (9)$$

where $\lambda_{1,\tau}$ and $\lambda_{2,\tau}$ are two tuning parameters that satisfy $\frac{\lambda_{2,\tau}}{\lambda_{1,\tau}+\lambda_{2,\tau}} > 0$. This restriction implies that the EN is strictly convex, so it forces high pairwise correlated predictors to have similar coefficients. As a result, EN stretches the net that retains all the important predictors, even if they are highly correlated.

As shown by Zou and Hastie (2005), we can reformulate the EN objective function as a LASSO problem[8]. This has computational appealing properties, as we can use the Belloni and Chernozhukov (2011) algorithm to estimate the EN. To implement this idea lets define (for simplicity assume that $\boldsymbol{X_t}$ contains all covariates): $y_t^+(\tau) = \left(y_t\ O_p\right)'$and $\boldsymbol{X_t^+}(\tau) = \frac{1}{\sqrt{1+\lambda_{2,\tau}}}\left(\boldsymbol{X_t}\ \sqrt{\lambda_{2,\tau}}\boldsymbol{I_p}\right)'$, where O_P represents a $p\ \times\ 1$ vector of zeros and $\boldsymbol{I_p}$ is a $p \times p$ identity matrix.

Based on this new formulation, Eq. (10) can be re-expressed as follows,

$$\min_{\boldsymbol{\phi}^{++}} E\left[\rho_\tau(y_t^+ - X_t^+\boldsymbol{\phi}(\tau))\right] + \gamma_\tau \sum_{i=1}^{p}|\phi(\tau)_i|, \tag{10}$$

where $\gamma_\tau = \frac{\lambda_{1,\tau}}{\sqrt{1+\lambda_{2,\tau}}}$. Notice that now the sample size is now equal to $T\ +\ p$, which enables EN to select all p high-frequency predictors in all cases. To remove the double shrinkage effect from LASSO, the EN-Q estimator is $\boldsymbol{\phi}^+(\tau) = (\mathbf{1} + \boldsymbol{\lambda_{2.\tau}})\boldsymbol{\phi}^{++}(\tau)$ (see Zou and Hastie, 2005). In our application, we only apply this correction if we use the EN-Q model to produce directly the conditional quantile nowcasts.

3.2.4. Soft and Hard thresholding: LASSO-PCA-Q and EN-PCA-Q

We consider the approach of soft and hard threshold methods applied to forecasting with many predictors, in the same vein that Lima et al. (2020) and Bai and Ng (2008). To implement this, first, we estimate principal components from the non-zero coefficients selected by LASSO-Q or EN-Q. Then, from these selected variables, we estimate factors by principal components analysis and select the optimal number of factors using the eigen ratio (Ahn and Horenstein, 2013). Finally, we keep the factors with associated p-values lower than 0.01 (or the statistically most significant one). To differentiate these models from those that only use the soft threshold to produce forecasts (i.e., LASSO-Q and EN-Q), we label these types of models as LASSO-PCA-Q and EN-PCA-Q, with the first step selected by LASSO or the elastic net, respectively.

3.2.5. Adaptive sparse group LASSO (ASGL-Q)

We introduce a novel framework based on an additive sparce group LASSO applied to quantile regressions (ASGL-Q), proposed by Mendez-Civieta et al. (2021)[9]. This technique is especially

[8] Bai and Ng (2008) and Lima and Godeiro (2020) use this approach to produce conditional mean forecasts with different loss functions. The former apply the mean square error and the latter the tick loss function.
[9] We thank an anonymous reviewer for suggesting this model.

suited for high-dimensional problems were $(p >> T)$ as for dealing with multiple groups of high-frequency variables, with sparsity allowed within the high-frequency lagged vector. Also, it uses adaptive weights in the penalization scheme, in line with Zou (2006). For notation simplicity, we refer to $TL_\tau(\boldsymbol{\phi})$ as the quantile function of all the vector of parameter $\boldsymbol{\phi}$ defined as in Eq. (2). The *ASGL-Q* objective function is given by:

$$\min_\phi E\,[TL_\tau(\boldsymbol{\phi}) + \alpha\lambda \sum_{j=1}^{K} \widetilde{w}_j \left|\phi(\tau)_j\right| + (1-\alpha)\lambda \sum_{l=1}^{M} \sqrt{p_l}\,\tilde{v}_l \|\phi(\tau)^l\|_2\,], \qquad (11)$$

where $\widetilde{w}_j$ is the weight for the *j-th* parameter $\phi(\tau)_j$ and $\tilde{v}_l$ is the weight for the *l-th* group of parameters (or high-frequency variable) $\phi(\tau)^l$. Overall, these weights assign a low weight to a relatively important high-frequency variable (or to a given lag), thus penalizing less. Notice that Eq. (11) is a linear combination of LASSO and group LASSO, given by the degree of penalization λ and the tradeoff between the two of them given by $\alpha \in [0,1]$. Specifically, a value close to 1 leads to additive LASSO and 0 additive sparse group LASSO. Thus, this formulation provides solutions that are both between and within group sparse. Also, as pointed out by Mendez-Civieta et al. (2021) this formulation defines a convex function which ensures that solution of the minimization process is a global minimum.

3.3. Forecast Combination

There is extensive literature that documents that forecast combinations provide a superior performance, as they use the information from all the underlying models rather than relying on one specific model (e.g., Stock and Watson, 2004; Andreou et al, 2013; Pettenuzzo et al. 2016; Ferrara et al., 2021). Indeed, selecting just one model can be misleading and highly inconvenient in the presence of misspecification (Hansen et al. 2011). While there are different methods for implementing forecast combinations, we rely on the discounted mean squared forecast error combinations approach (Stock and Watson, 2004; Andreou et al, 2013), using the tick loss as the objective function.

The weights are computed recursively in the following way:

$$w_{i,t-h_d} = \frac{\lambda_{i,t-h_d}^{-1}}{\sum_i^N \lambda_{i,t-h_d}^{-1}}, \quad \lambda_{i,t-h_d} = \sum_{s=T_o}^{T_f} \delta^{T_f-s}\left(y_t - GaR_{i,s}\right) * \left(\tau - \mathbb{1}_{y_t < GaR_{i,s}\,(10\%)}\right), \quad (12)$$

where $w_{i,t-h_d}$ is the weight of model i, which depends on the discounted tick loss given by $\lambda_{i,t-h_d}$, with discount factor $\delta = 0.9$, and where T_o is the point at which the first prediction is

computed, and T_f is the point at which the most recent prediction can be evaluated with the high-frequency indicator up to the latest available observation.

3.4. GaR evaluation

We evaluate tail risk forecasts by using a battery of indicators developed in the literature on forecast and risk management. Our main tool for assessing GaR point forecasts is the average tick loss, which has been shown to be particularly suitable when the object of interest is the forecast of a certain quantile of the dependent variable's conditional distribution (see Giacomini and Komunjer, 2005, Gneiting and Raftery 2007; Gneiting and Ranjan 2011, Manzan, 2015). In particular, Carriero et al. (2020) use it to evaluate the predictive capacity of their models for quantifying tail risks.

The average tick loss for the $\tau = 0.10$ is specified as follows,

$$TL_{0.10} = \frac{1}{T}\sum_{t=1}^{T}(y_t - GaR_t\ (10\%)) * (\tau - \mathbb{1}_{y_t < GaR_t\ (10\%)}), \qquad (13)$$

where $GaR\ (10\%)$ refers to the 10% quantile of GDP growth conditioned by the information set available up to T for the given model and the high-frequency predictor. The indicator function $\mathbb{1}_{(y_t < GaR_t\ (10\%))}$ takes a value of 1 if it is below the 10% forecast quantile and 0 otherwise. To assess the statistical significance, we estimate Diebold and Mariano (1995) t-tests for equality predictability of models, with the loss function defined as the tick loss.

In addition, we employ tests commonly used in the risk management literature to assess interval forecasts. Following Christoffersen (1998), the problem of assessing the adequacy of a Value-at-Risk (VaR) model can be reduced to the problem of determining whether the indicator of excess sequence (i.e., the $\mathbb{1}_{(y_t < GaR_t\ (10\%))}$) has two properties: i) an unconditional coverage property, and ii) an independence property. In this setting, GaR forecasts are evaluated using the TL, which is a loss function generally used to assess the accuracy of VaR predictions (Giacomini and Komunjer, 2005). We evaluate these two conditions with the Unconditional Coverage (UC), and the Dynamic Quantile (DQ) tests (Engle and Manganelli, 2004), respectively. Specifically, the DQ is estimated using four lags of the excess sequence indicator (see Engle and Manganelli, 2004). Brownlees and Souza (2021) follow a similar approach for a multi-country GaR evaluation. Note that these two criteria can be achieved by more than one model, so at the end the TL is used in the final selection of the best performing model.

4. Empirical analysis

This section presents the statical details for each model's computation and the results of the nowcasting exercise.

4.1. Parametrization and computational approach

In the empirical analysis that follows, we consider different high-frequency GaR measures spanning different variables and models. Below we discuss the choice of our parameters for each model estimated recursively from 2006Q1 to 2020Q4.

- MIDAS-Q: Eq. (6) is estimated through a quantile approach described in Section 3.1 considering a third degree Almon lag polynomial ($c = 3$) with two end-point restrictions r=2, such that the parameters estimation is reduced substantially to $K = c - r + 2$ (including the constant and the lagged term). This lag structure has nice economic properties as it slowly decays towards zero (see Mogliani and Simoni, 2021).
- BMIDAS-Q: Based on the abovementioned constrained Almon lag structure for the high-frequency variable, Eq. (7) is estimated through the Bayesian methodology considered in Section 3.2.2. Specifically, the model considers standard uninformative priors on the coefficient vector to have a mean of 0 and a variance that takes a diagonal equal to 9, except for the autoregressive lag of GDP, whose prior mean and variance are set to 0.5 and 0.1, respectively. The scale and shape parameters of the inverse gamma function are set to 0.01. The Gibbs sampler is used to estimate the model parameters with 10,000 repetitions (for computation efficiency), after a burn-in period of 1000 iterations, using the normal approximation, which simplifies the algorithm (Yang et al., 2015)[10]. The choice of these parameters closely resembles the ones of Ferrara et al. (2021), which is a natural benchmark model for our work.
- LASSO-Q: Following the model presented in Section 3.2.3, we set the penalty parameter λ equal to the 0.9 quantile of the pivotal distribution (Belloni and Chernozhukov, 2011). Figure A1 shows the selected lags for LASSO-Q using each high-frequency predictor. Interestingly, historically LASSO not only tends to select the most current daily lag of the given quarter, as one could expect, but others from past quarters. This is a key

[10] See Kozumi and Kobayashi (2011) for details on the estimation procedure.

difference of this technique compared to MIDAS-Q and BMIDAS-Q, as the latter has an arbitrary decaying weighting structure.

- EN-Q: Based on the model presented in Section 3.2.4, $\lambda_{1,\tau}$ is set as the penalty parameter of LASSO-Q model, defined as above, and $\lambda_{2,\tau}$ is obtained by minimizing the mean cross-validated errors of the model, with the EN mixing parameter set to $\alpha = 0.5$ (Friedman et al., 2010). Figure A2 shows the selected lags for EN-Q using each high-frequency predictor. Analogous to LASSO-Q, we observe that this model historically selects different daily lags for each high-frequency variable.
- LASSO-PCA-Q and EN-PCA-Q: Based on the non-zero high-frequency lags selected with either LASSO-Q or EN-Q, we estimate factors by principal components analysis, select the optimal number of factors using the eigen ratio (Ahn and Horenstein, 2013), and keep the factors with associated p-values lower than 0.01 (or the statistically most significant one). The final step of this procedure is estimated through the quantile approach described in Section 3.1.
- ASGL-Q: Based on the model presented in Section 3.2.6, we consider the following parametrization procedure. First, we carry out cross validation over different values of lambda and alpha to get their optimal values. By estimating this model with the full sample and all the high-frequency variables (except CISS, which is used as benchmark), we obtain the optimal values of $\lambda = 0.010$ and $\alpha = 0.25$, which minimize the tick loss function. Second, we compute recursively both LASSO weights and group LASSO weights based on regression on a subset of principal components. As suggested by Mendez-Civieta et al. (2021) this method achieves better results in terms of prediction error and stability of the variables selected when used in real datasets.

Subsequently, for each model we use the forecast combination from Section 3.3 to summarize the information steaming from different daily indicators. Specifically, we implement the following procedure.

- For each model considered, the individual GaR for each X_i, i.e. the conditional 10% quantile $\widehat{Q_{0.10}}(y_T|y_{T-h}, \boldsymbol{X_i})$, is computed recursively on a daily basis starting from January 1, 2007 to December 31, 2020, where X_i stands for the high-frequency variable. In the case of the AGSL-Q model, since it uses all high-frequency variables, it produces directly a combined-GaR prediction.

- Then, using the combination weights calculated as in Eq. (12) we compute the combined GaR recursively for each model (except for the ASGL-Q) as follows:

$$\widehat{GaR}_T^* = \sum_i w_{i,T-h_d} * \widehat{Q_{0.10}}(y_T|y_{T-h}, \boldsymbol{X_i})$$

This formula allows to capture the relative importance of individual GaR model and to deal with the potential problem of introducing many series, potentially correlated, into a common framework. It is important to mention that in this formulation we do not include the CISS since it is used as the benchmark indicator. Figure A3 provides a recursive plot of the combination weights assigned to the various models using forecast combination. We can observe that both real and financial indicators are important to produce accurate GaR forecasts and the importance of each one is time varying.

4.2. Nowcasting GaR

We recursively estimate all the above mentioned specifications for each quarter from 1986Q1 to 2006Q4 and construct daily GaR nowcast in pseudo-real-time starting from January 1, 2007.

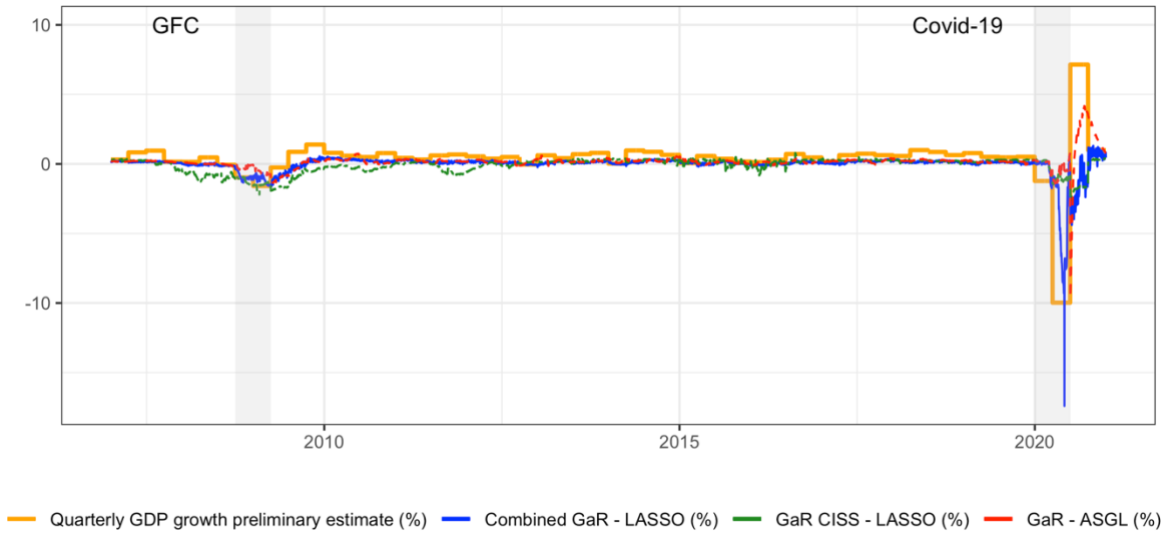
As a starting point, we show the combined-GaR forecasts of our LASSO-Q model[11] compared to two alternative specifications, the individual GaR using the CISS and the combined-GaR estimated by ASGL-Q. Recall that while LASSO-Q uses forecast combination as explained in Section 3.3, ASGL-Q uses all the information directly in the estimation. Figure 1a shows the preliminary estimates in real time of the quarterly growth rate of the US along with combined GaR and the two benchmark models. Overall, we observe that the large negative growth rates that appear during recession periods, such as the Global Financial Crisis (GFC) in 2008–2009 and the Covid-19 pandemic (that started in 2020), are both captured effectively by our combined-GaR, while it is not the case for last event for the alternative specifications. First, the difference in prediction between LASSO-Q and ASGL-Q are partially due to the difference in their combination schemes (see Figure 1b and 1c). While the weights of the former model are more volatile, those of the latter are more stable. Interestingly, in both frameworks, at the onset of the Covid-19 pandemic more weight is given to the ADS indicator, highlighting the usefulness of using real indicators in a GaR framework. Second, relative to the individual GaR model using the CISS, a key aspect of our framework is that it allows both financial and real indicators to

[11] We chose to show the combined-GaR nowcasts of the LASSO-Q model as, in general, it outperforms the rest of the models in terms of the average tick loss (see Table 2). Results for the other models are available upon request.

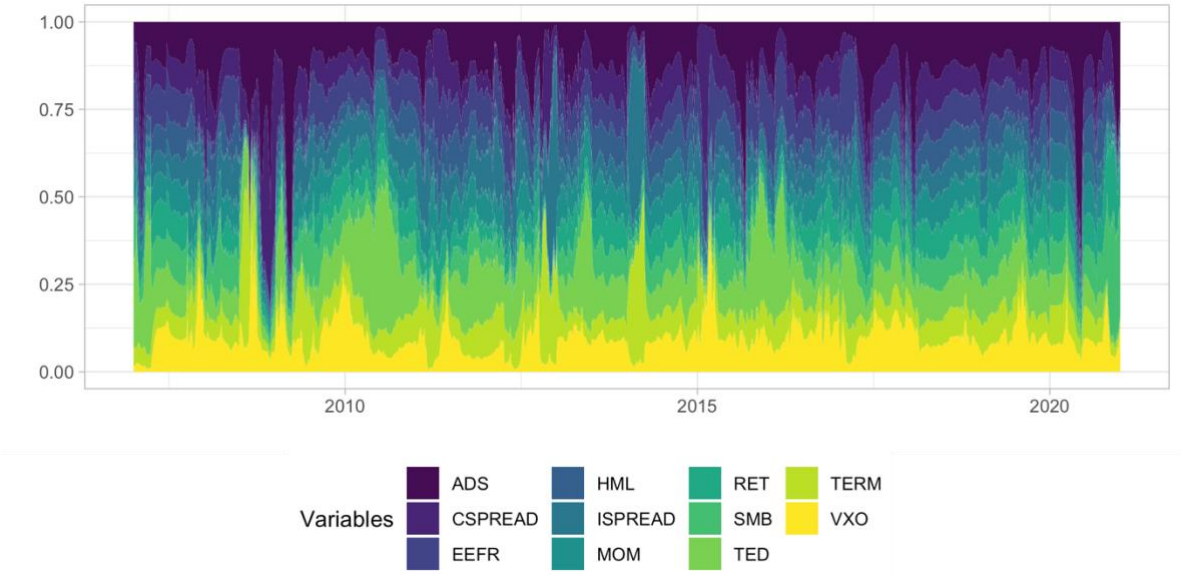
improve the accuracy of GaR nowcasts using the combination of forecasts, unlike the standard GaR framework that uses only a composite financial condition index (Adrian et al., 2019). Figure 1b provides a clearer view of this point by presenting a recursive plot of the combination weights assigned to the different individual GaR nowcasts. We can observe that the relative importance of real and financial indicators is time varying. Indeed, we observe that around the global financial crisis that started in 2008, the ADS, VXO and credit spread (CSPREAD) indicators receive a relatively high weight across all models. In contrast, on the onset of the Covid-19 pandemic, all models assign higher weights to the ADS. On one hand, the former is in agreement with the general consensus in the macro-financial literature that highlights the financial nature of the GFC, in which financial intermediaries and markets amplified the shocks to the real economy (Isohätälä et al., 2016; Brunnermeier and Sannikov, 2016; Gertler and Gilchrist, 2018). On the other hand, the latter results recognize that the Covid-19 crisis was a product of the supply restrictions imposed to contain the pandemic, which were real and supply-side in nature (Guerrieri et al., forthcoming). Consequently, our estimated optimal weights suggest that it is fundamental to include real and financial daily indicators to improve GaR nowcasts. Interestingly, related works by Ferrara et al. (2021) and De Santis and Van der Veken (2020), which only include daily financial variables in the GaR framework, fail to capture the real magnitude of the risks during the Covid-19 period. This evidence indicates that financial variables alone play only a modest role in gauging the effect of this last recession.

Figure 1. GaR results for LASSO-Q and AGLS-Q

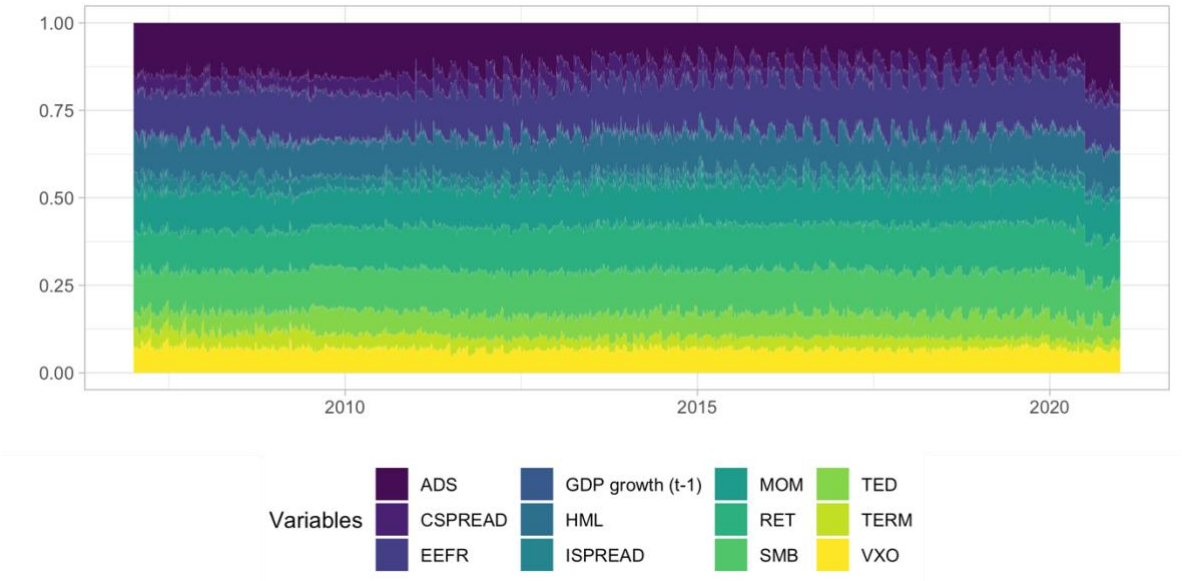
a. GaR nowcasts



b. Daily combination weights for LASSO



c. Daily group weights for ASGL-Q



Sources: FRED database and authors' computations.
Note: Time span 1986Q1 to 2020Q4. The gray shaded area represents NBER recessions at the end of the period.

4.3. Evaluation

In this section we perform the following exercises to assess the relative performance of: i) combined GaR nowcasts based on real and financial indicators compared to individual GaR nowcasts using a financial condition index (CISS) as a benchmark; ii) individual GaR nowcasts of financial and real indicators relative to the combined GaR nowcasts, in the spirit of Figueres and Jarocinski (2020).

Combined GaR (using financial and real variables) versus standard GaR

Table 2 reports the relative average tick loss of a given combined GaR model compared to the benchmark model (and individual GaR using the CISS), joint with its Diebold-Mariano (DM) probability test. A value lower than one in the TL implies that the combined model outperforms the benchmark, while for the DM the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred). Notably, the vast majority of models outperform the benchmark and rejects the hypothesis of equality of forecasts according to the DM test, which gives strong evidence of the benefits of combining multiple real and financial indicators into the GaR framework. Remarkably, LASSO-Q model tends to provide lower average tick loss and often rejects the null hypothesis of the DM test for different daily horizons, for both periods (including and not the covid-19). We provide evidence that LASSO

lag selection improves forecast accuracy while imposing fewer restrictions than traditional MIDAS models.

Table 2. Out-of-sample results: Relative tick loss of combined-GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR-MIDAS	0.641	0.001	0.655	0.001	0.653	0.000	0.686	0.001	0.683	0.001
GaR-BMIDAS	0.606	0.000	0.616	0.000	0.631	0.001	**0.643**	**0.001**	**0.654**	**0.001**
GaR-LASSO	**0.59**	**0.001**	**0.559**	**0.000**	**0.569**	**0.000**	0.769	0.145	0.843	0.232
GaR-EN	0.956	0.415	0.978	0.461	0.932	0.366	0.853	0.273	0.858	0.277
GaR-LASSO-PCA	0.617	0.001	0.638	0.002	0.706	0.01	0.83	0.225	0.857	0.266
GaR-EN-PCA	0.617	0.001	0.691	0.01	0.741	0.039	0.809	0.176	0.844	0.251
GaR-ASGL	1.102	0.646	1.037	0.559	0.983	0.471	0.945	0.419	1.221	0.744
Including COVID-19 (2007Q1 to 2020Q4)										
GaR-MIDAS	**0.855**	**0.027**	0.82	0.005	0.804	0.022	0.558	0.094	0.943	0.201
GaR-BMIDAS	0.878	0.021	0.849	0.000	0.839	0.005	0.558	0.087	0.932	0.141
GaR-LASSO	0.864	0.002	0.773	0.006	**0.458**	**0.096**	**0.501**	**0.121**	**0.895**	**0.092**
GaR-EN	0.953	0.243	0.969	0.33	0.822	0.153	0.563	0.139	0.917	0.173
GaR-LASSO-PCA	0.94	0.263	**0.733**	**0.041**	0.488	0.116	0.593	0.133	0.927	0.12
GaR-EN-PCA	0.911	0.102	0.85	0.013	0.841	0.064	0.691	0.116	0.903	0.123
GaR-ASGL	1.106	0.79	1.002	0.506	1.027	0.614	1.056	0.687	1.085	0.766

Note: This table shows the tick loss for each combined GaR relative to the individual GaR considering the CISS, for different daily horizons. Bold values represent the best model across the models considered for each daily horizon h_d. Diebold and Mariano t-test (DM) to test equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred).

Table 3 reports the tests commonly used in the financial risk management literature, namely the UC and the DQ tests, to assess interval forecasts for the different combined GaR models. Specifically, the UC test reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the DQ test the probability that the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Overall, models that use LASSO or elastic net do it better than MIDAS in this adequacy test,

with higher number of not rejections at 10% probability. These results hold for both, the period before Covid-19 and the period including it. The results hold for alternative models presented in Appendix B.

Table 3. Out-of-sample results: Backtesting tests of combined-GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR-MIDAS	0.001	**0.619**	0.019	0.164	0.019	**0.144**	0.019	**0.141**	0.001	**0.619**
GaR-BMIDAS	0.001	**0.619**	0.001	0.619	0.001	**0.619**	0.001	**0.619**	0.001	**0.619**
GaR-LASSO	0.019	**0.849**	**0.273**	0.014	**0.273**	0.018	0.095	**0.59**	0.019	**0.272**
GaR-EN	**0.273**	**0.18**	**0.273**	**0.218**	**0.926**	**0.126**	**0.273**	0.045	**0.273**	**0.107**
GaR-LASSO-PCA	0.095	**0.316**	0.095	**0.344**	**0.273**	**0.378**	0.095	**0.63**	0.095	0.011
GaR-EN-PCA	0.019	**0.842**	**0.565**	0.021	**0.273**	**0.386**	0.095	**0.603**	0.095	**0.631**
GaR-ASGL	0.427	0.013	0.226	0.071	0.226	0.044	**0.926**	**0.2**	**0.226**	**0.493**
Including COVID-19 (2007Q1 to 2020Q4)										
GaR-MIDAS	**0.208**	0.001	0.455	0.045	**0.455**	0.085	**0.455**	0.024	**0.208**	0.003
GaR-BMIDAS	0.068	0.04	0.068	0.063	0.068	0.08	0.068	0.08	0.068	0.08
GaR-LASSO	0.068	**0.917**	**0.786**	0.000	**0.786**	0.021	**0.786**	0.042	**0.208**	0.266
GaR-EN	**0.786**	0.007	**0.786**	0.009	**0.547**	0.029	**0.786**	0.042	**0.786**	0.076
GaR-LASSO-PCA	**0.455**	**0.235**	**0.455**	0.009	**0.786**	0.036	**0.455**	**0.118**	**0.455**	0.008
GaR-EN-PCA	**0.208**	**0.468**	**0.547**	0.000	**0.86**	**0.202**	**0.455**	**0.13**	**0.455**	**0.677**
GaR-ASGL	0.16	0.015	0.031	0.000	0.031	0.000	**0.312**	**0.237**	**0.031**	**0.371**

Note: This table shows the following two backtesting tests for different combined GaR nowcasts: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), in which the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Bold values indicate which model passes these two tests with 10% probability.

Combined-GaR (using financial and real variables) versus individual GaR

Now we address the question whether combined or individual indicators provide more accurate nowcasts, in the spirit of Figueres and Jarocinski (2020). Table 4 reports the relative average tick loss of a given individual GaR nowcasts relative to the combined GaR nowcasts (benchmark)

using LASSO-Q, joint with its Diebold-Mariano (DM) probability test. Again, a value lower than one in the TL implies that the individual GaR outperforms the benchmark, while for the DM the alternative hypothesis is that the indicated forecast is more accurate than the benchmark. In this setting, we would prefer to have a tick loss greater than 1 and not reject the null hypothesis, to have evidence of our combined GaR framework. All in all, results suggest that with 10% probability we cannot reject the null hypothesis of the DM test, meaning that combined GaR nowcasts more adequate information. Nonetheless, the individual GaR specification using the ADS indicator is the only specification which has a relative TL lower than one for daily horizons greater than 10 and for the period including the Covid-19 period. These results suggest that individual GaR models that include the ADS outperform those that do not include it, as in Pettenuzzo *et al.* (2016) and Lima et al. (2020). Also, this result recognize that the Covid-19 crisis was a product of the supply restrictions imposed to contain the pandemic, which were real and supply-side in nature (Guerrieri et al., forthcoming). Besides this, we find that financial conditions indicators, such as the CISS, provide often more accurate forecasts than individual indicators, as suggested by other works (Adrian et al., 2019; Figueres and Jarocinski (2020). The results hold for alternative models presented in Appendix C.

Table 4. Out-of-sample results of LASSO-Q: Relative tick loss of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	2.232	0.978	2.072	0.99	1.768	0.988	1.772	0.997	1.692	0.996
GaR-EEFR	2	0.965	2.392	0.935	1.819	0.972	1.711	0.971	1.649	0.968
GaR-RET	2.107	0.973	2.049	0.962	1.792	0.946	2.2	0.994	1.594	0.971
GaR-SMB	2.079	0.991	2.202	0.991	1.833	0.994	1.93	0.986	2.069	0.99
GaR-HML	2.2	0.957	1.447	0.916	1.57	0.973	1.514	0.968	1.655	0.916
GaR-MOM	**1.448**	**0.987**	2.076	0.959	1.578	0.987	2.072	0.96	1.756	0.973
GaR-VXO	1.813	0.991	1.55	0.995	1.521	0.997	1.258	0.921	1.268	0.886
GaR-CSPREAD	2.242	0.979	1.789	0.993	1.367	1	1.275	0.963	1.259	0.951
GaR-TERM	2.198	0.984	1.946	0.991	1.728	0.989	1.754	0.99	1.684	0.989
GaR-TED	1.643	0.918	1.603	0.968	1.485	0.964	1.44	0.96	1.397	0.931
GaR-ADS	1.678	0.95	1.496	0.91	**1.175**	**0.717**	**0.939**	**0.375**	1.263	0.915
GaR-CISS	1.62	0.999	**1.448**	**0.99**	1.349	0.961	1.236	0.824	**1.185**	**0.749**
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	1.445	0.968	1.555	0.995	1.53	0.99	1.837	0.973	1.346	0.983

GaR-EEFR	1.449	0.982	1.707	0.992	1.55	0.987	1.855	0.963	1.367	0.979
GaR-RET	1.408	0.992	1.556	0.989	1.503	0.976	1.776	0.964	1.208	0.957
GaR-SMB	1.271	0.948	1.51	0.989	1.301	0.991	1.667	0.96	1.302	0.996
GaR-HML	1.453	0.995	1.504	0.983	1.281	0.913	1.834	0.942	1.3	0.975
GaR-MOM	1.274	0.981	1.712	0.971	1.51	0.969	1.722	0.934	1.307	0.988
GaR-VXO	1.196	0.908	1.335	0.995	1.317	0.994	1.426	0.893	1.129	0.95
GaR-CSPREAD	1.336	0.991	1.351	0.993	1.28	0.939	1.501	0.888	1.133	0.956
GaR-TERM	1.42	0.974	1.502	0.995	1.432	0.993	1.789	0.971	1.334	0.989
GaR-TED	1.315	0.96	1.433	0.994	1.42	0.985	1.731	0.95	1.279	0.98
GaR-ADS	1.375	0.923	**0.595**	**0.174**	**0.655**	**0.159**	**0.504**	**0.152**	**0.743**	**0.26**
GaR-CISS	**1.098**	**0.898**	1.177	0.987	1.188	0.936	1.446	0.884	1.108	0.877

Sources: FRED database and authors' computations.

Note: This table shows the tick loss for each individual GaR relative to the combined GaR for different daily horizons. Bold values represent the best model across the models considered for each daily horizon h_d. Diebold and Mariano t-test (DM) to test equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark.

Table 5 reports the UC and the DQ tests, to assess interval forecasts for the different individual GaR specifications estimated by LASSO-Q. Again, the UC test reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the DQ test the probability that the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Remarkably the individual GaR that include the CISS pass all the adequacy tests for all the daily horizons with 10% probability, including or not the Covid-19 period. For the rest of the indicators, the evidence is mixed and depends on the daily horizon.

Table 5. Out-of-sample results for LASSO-Q: UC and DQ tests of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	**0.926**	**0.161**	**0.926**	0.072	**0.926**	**0.121**	**0.565**	0.07	**0.565**	**0.052**
GaR-EEFR	**0.717**	**0.131**	**0.226**	0.012	**0.427**	0.07	**0.226**	0.089	**0.107**	0.014
GaR-RET	**0.226**	0.001	0.001	0.000	0.046	0.000	0.046	0.000	0.001	0.000
GaR-SMB	0.046	0.001	**0.427**	0.015	**0.226**	**0.103**	0.006	0.001	**0.107**	0.002
GaR-HML	0.046	0.05	**0.107**	0.000	**0.226**	0.001	0.018	0.006	0.018	0.003
GaR-MOM	**0.427**	**0.781**	**0.717**	**0.146**	0.046	0.099	0.046	0.000	**0.226**	0.003
GaR-VXO	**0.107**	0.066	0.046	**0.812**	0.018	0.063	0.046	0.067	0.046	**0.707**

GaR-CSPREAD	**0.226**	0.036	**0.226**	0.038	**0.717**	**0.741**	**0.717**	**0.799**	**0.565**	**0.921**
GaR-TERM	**0.565**	0.087	**0.565**	0.088	**0.565**	0.08	**0.926**	**0.28**	**0.273**	**0.701**
GaR-TED	**0.226**	0.011	**0.107**	0.003	**0.107**	0.003	0.046	0.003	0.046	0.023
GaR-ADS	**0.107**	0.067	0.018	0.05	**0.107**	0.043	**0.226**	**0.186**	**0.226**	**0.23**
GaR-CISS	**0.926**	**0.907**	**0.717**	**0.875**	**0.926**	**0.861**	**0.427**	**0.871**	**0.273**	**0.878**
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	**0.312**	0.003	**0.312**	0.002	**0.312**	0.002	**0.547**	0.001	**0.547**	0.001
GaR-EEFR	**0.16**	0.007	0.031	0.000	0.074	0.001	0.031	0.002	0.012	0.000
GaR-RET	0.074	0.000	0.000	0.000	0.004	0.000	0.004	0.000	0.000	0.000
GaR-SMB	0.012	0.001	**0.16**	0.004	0.074	0.083	0.001	0.000	0.031	0.001
GaR-HML	0.004	0.000	0.012	0.000	0.031	0.001	0.001	0.000	0.004	0.002
GaR-MOM	**0.16**	**0.425**	**0.16**	0.018	0.004	0.001	0.004	0.000	0.074	0.000
GaR-VXO	0.031	**0.338**	0.004	**0.267**	0.001	0.011	0.004	0.042	0.004	**0.215**
GaR-CSPREAD	0.031	0.005	0.031	0.011	**0.16**	**0.204**	**0.16**	**0.427**	**0.547**	**0.921**
GaR-TERM	**0.547**	0.002	**0.547**	0.001	**0.547**	0.002	**0.312**	0.008	**0.86**	0.021
GaR-TED	0.031	0.003	0.012	0.000	0.012	0.000	0.004	0.000	0.004	0.001
GaR-ADS	0.074	**0.137**	0.004	0.003	0.031	0.001	0.074	**0.23**	0.031	0.018
GaR-CISS	**0.547**	**0.802**	**0.312**	**0.795**	**0.547**	**0.807**	**0.16**	**0.71**	**0.786**	**0.885**

Sources: FRED database and authors' computations.

Note: This table shows the following two backtesting tests for different individual GaR nowcasts: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), in which the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Bold values indicate which model passes these two tests with 10% probability.

Overall, the evidence we present here supports the time-varying importance of both daily financial and real indicators for estimating GaR. Our results are consistent with those for the Euro Zone reported by Ferrara et al. (2020) and for the US reported by De Santis and Van der Veken (2020), in that financial variables provide policymakers with timely warnings about the downside risks of GDP. Nevertheless, we provide further and clearer evidence in the way suggested by Pettenuzzo et al. (2016), emphasizing the usefulness of incorporating a real high-frequency indicator in the forecasting regressions, such as the ADS. Furthermore, by comparing our combined GaR models against the standard GaR using a financial condition index, namely the CISS, we show that our framework performs significantly better. This is also true when comparing our combined-GaR framework against individual GaR models, although the ones that only include the ADS perform similarly when considered the Covid-19 period. All in all,

this evidence ultimately suggests that financial variables alone played a limited role in gauging the downside risk for the GDP of the Covid-19 pandemic, and acknowledges the complex ways in which real and financial variables interconnect and determine economic growth, which is causal in nature.

5. Conclusions

We show that on a daily frequency both real and financial variables provide valuable information for monitoring vulnerable periods of economic activity. Our key contribution is that we show that by incorporating these two types of variables simultaneously in the GaR framework, we can provide an early signal of GDP downturns in pseudo real-time that works well for both, the GFC and the Covid-19 episodes.

Our flexible approach allows us to emphasize the important role of economic theory and economic intuition when the results of forecast combination exercises are interpreted and also to improve the point forecast itself. By acknowledging the complexity of the nowcasting task in macroeconomics, especially when using high frequency data, we contribute to a better understanding of the economic signals that can be extracted from this daily information to anticipate economic declines. In this sense, we show that during the GFC and the Covid-19 pandemic, the optimal forecasting weights of real and financial variables clearly changed. In the former period, financial indicators were fundamental, such as credit spreads and the VXO; however, they failed to capture the magnitude of the GDP decline observed during the Covid-19 quarters. This, in turn, is explained by the different natures of the two crises, that we can only grasp because we understand (up to some extent) the economic mechanisms behind these two crises (i.e. their causes).

Interestingly, within the set of financial variables, VXO and CSPREAD are especially relevant across models in around the GFC, which highlights the prominent role of uncertainty in determining economic outcomes. However, as mentioned above, the financial indicators alone were unable to forecast GDP low quantiles during Covid-19. Indeed, only by including the ADS index we managed to gauge both the sign and the magnitude of the downside GDP risk in this period.

We showed that our combined-GaR model outperforms the standard GaR framework, which considers only financial condition indicators (Adrian et al., 2019; Ferrara et al. 2021). More technically, we have assessed this by: i) comparing the performance of combined GaR nowcasts

to individual GaR nowcasts using the CISS as a benchmark; and ii) individual GaR nowcasts of financial and real indicators relative to the combined GaR nowcasts (see Figueres and Jarocinski, 2020). We have implemented this exercise by using different dimension reduction techniques, namely MIDAS and shrinkage methods, and different tests based on the tick loss. Specifically, we have compared seven different models and 12 high frequency predictors using a forecast combination approach with time-varying optimal weights. Also, we have used a novel approach based on adaptive sparse group LASSO for quantile regressions, which allows for multiple groups and sparsity within the high-frequency lagged vector (Mendez-Civieta et al., 2021). While this model has some nice properties for high-dimensional problems, we found that LASSO-Q tend to outperform the rest of models in terms of forecast accuracy at different daily horizons. This is probably a consequence of traditional MIDAS restrictions on the lag structure of the high frequency indicator, which do not necessarily improve forecast accuracy. Therefore, our results give further support to past evidence, in that shrinkage models should ideally be used to select the number of lags of the high-frequency predictors. Regarding the ASGL-Q model, it displays more stable weights than LASSO-Q, which could arguably be a potential cause of the difference in the GaR nowcasts. Nonetheless, these two weighting schemes both suggests the forecasting importance of the ADS indicator during the Covid-19 period.

We used a single indicator for capturing the role of real economic activity, namely the ADS, basically because it is the only indicator that exists at the daily frequency. We also introduced weakly vintages of this indicator to perform to nowcasting exercise in pseudo-real time. Nevertheless, we believe that more indicators gauging the informational content of different facets of economic activity and the credit markets will prove to be fundamental in the future, not only to achieve greater forecasting accuracy in real time, but also to understand the causes of ongoing crises, even before the true causal mechanisms are clear to the professionals. In this sense, we see our models as a first step in the direction of anticipating and understanding economic dangers while they are occurring.

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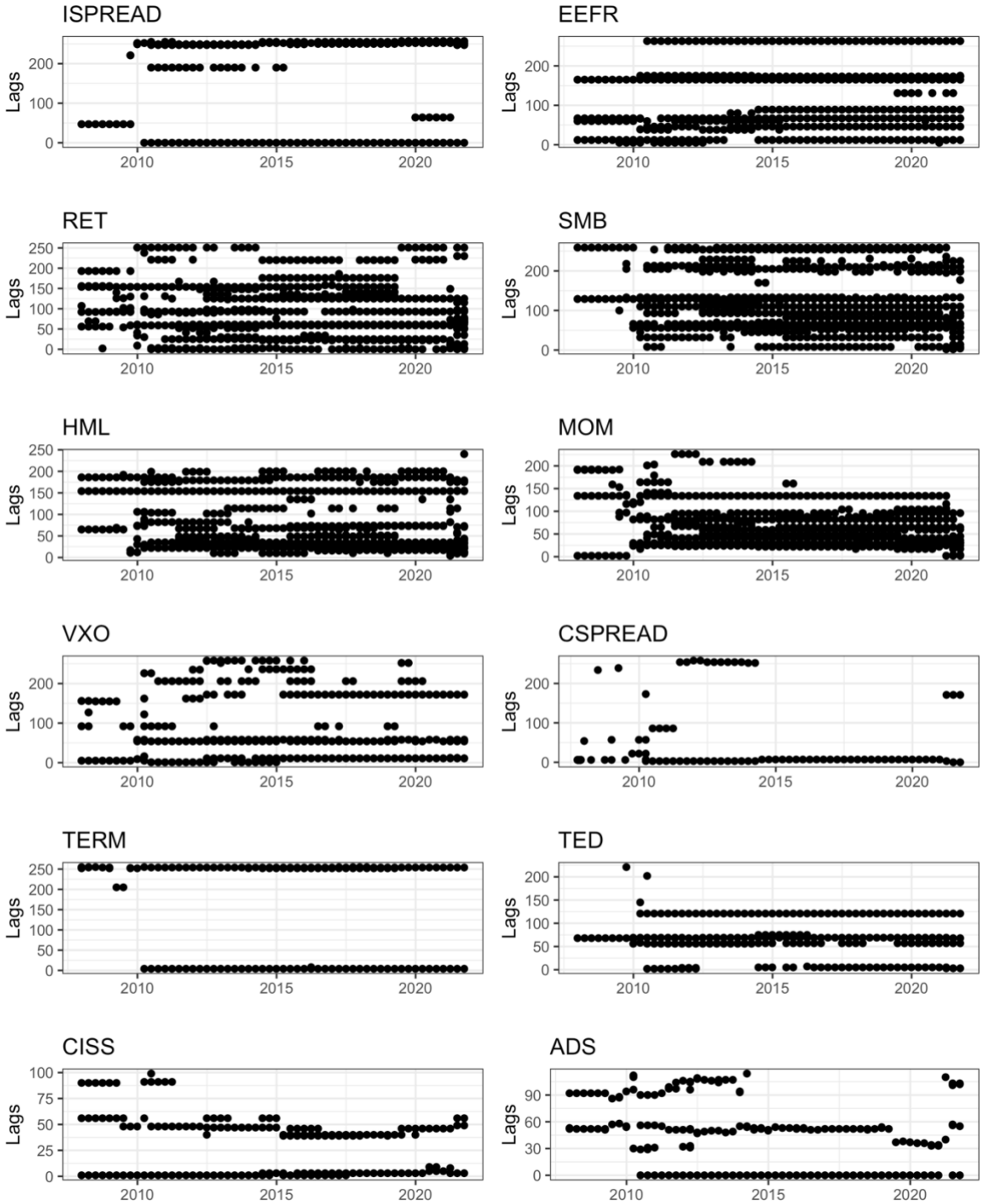
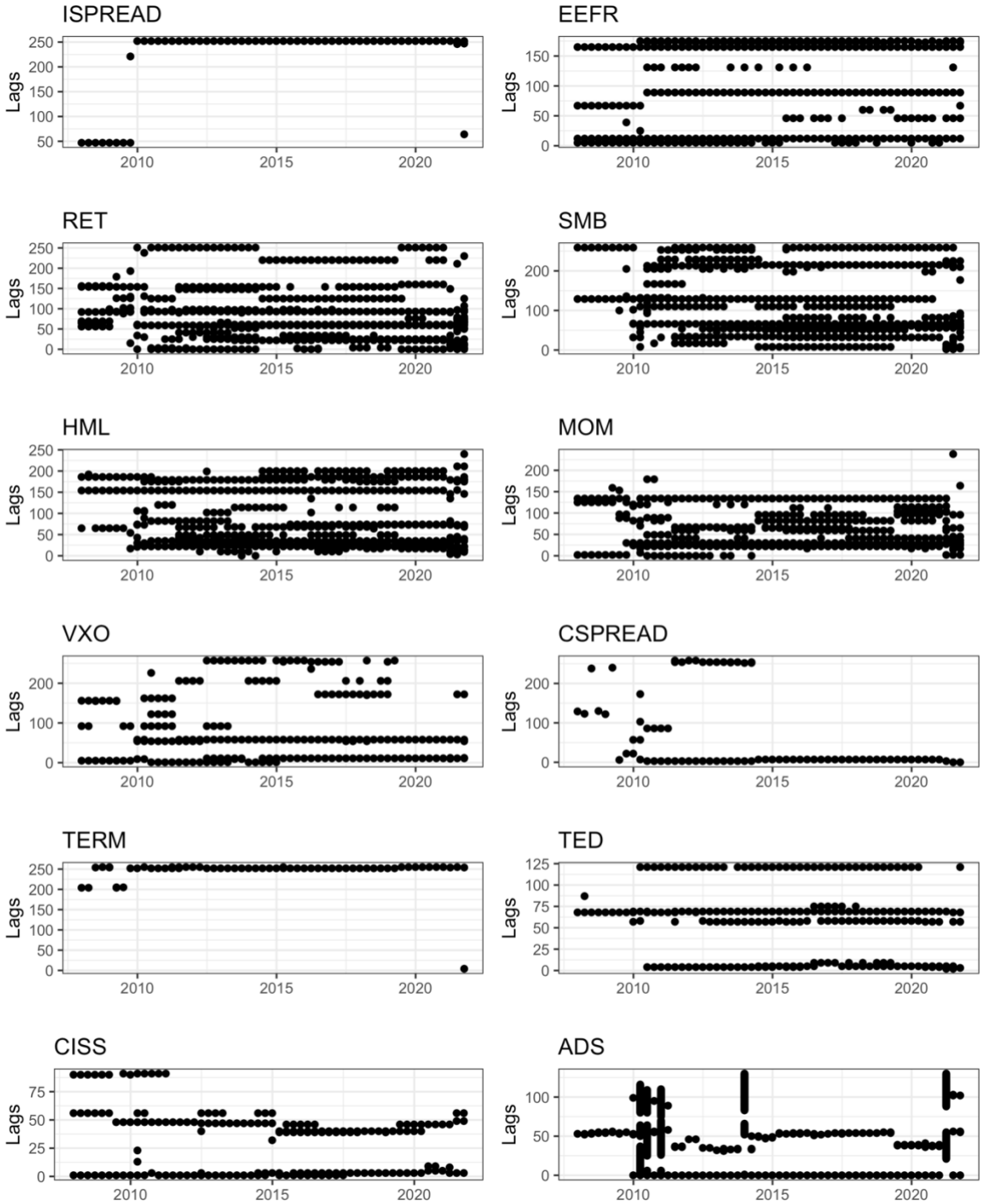
Appendix A

Figure A1. LASSO-Q selected high-frequency lags from individual models



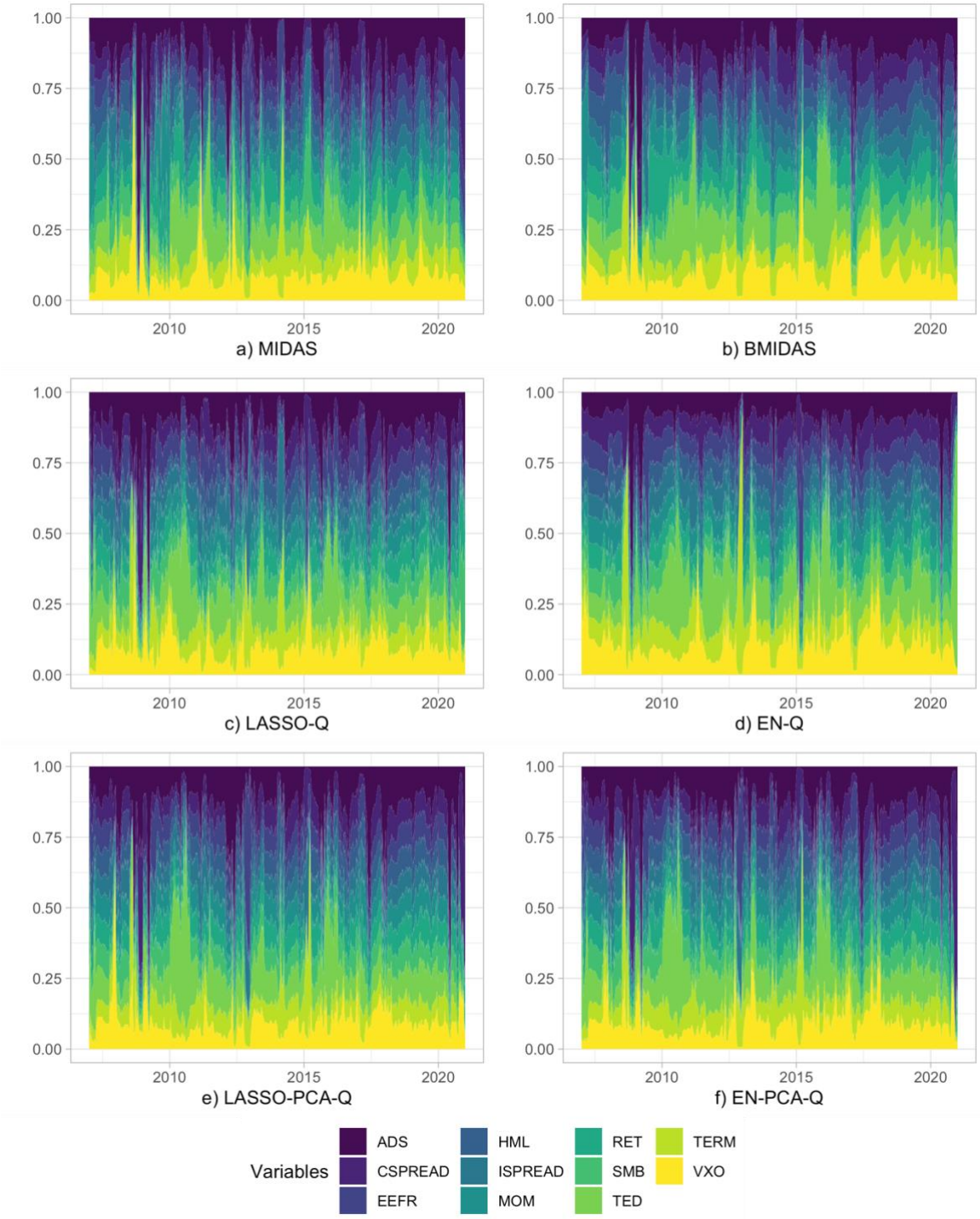
Source: Own elaboration.
Note: Figure shows the selected daily lags from LASSO-Q individual models that corresponds to the last day of each quarter.

Figure A2. EN-Q selected high-frequency lags from individual models



Source: Own elaboration.
Note: Figure shows the selected daily lags from EN-Q individual models that corresponds to the last day of each quarter.

Figure A3. Daily forecast combination weights



Source: Own elaboration.
Note: This figure plots the daily optimal weights for different GaR models.

Appendix B: TL for alternative models

Table B1. Out-of-sample results of MIDAS-Q: Relative tick loss of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	2.338	0.99	2.285	0.99	2.313	0.99	2.212	0.991	2.234	0.987
GaR-EEFR	2.317	0.971	2.185	0.962	2.221	0.967	2.12	0.968	1.999	0.955
GaR-RET	1.446	0.989	1.506	0.994	1.595	0.997	1.808	0.996	1.637	0.996
GaR-SMB	2.11	0.988	2.124	0.991	2.07	0.993	2.008	0.992	2.058	0.989
GaR-HML	2.141	0.987	2.204	0.986	2.29	0.978	2.452	0.98	2.169	0.981
GaR-MOM	2.097	0.978	1.987	0.978	1.945	0.979	1.752	0.978	1.78	0.974
GaR-VXO	1.617	0.951	1.589	0.959	1.537	0.975	1.493	0.986	1.467	0.994
GaR-CSPREAD	2.052	0.996	1.834	0.997	1.579	1	1.363	0.997	1.414	1
GaR-TERM	2.313	0.989	2.253	0.989	2.252	0.988	2.173	0.991	2.198	0.987
GaR-TED	1.957	0.982	1.929	0.978	1.889	0.977	1.758	0.98	1.784	0.981
GaR-ADS	1.474	0.936	1.527	0.953	1.377	0.928	1.1	0.698	1.433	0.908
GaR-CISS	1.559	0.999	1.527	0.999	1.531	1	1.458	0.999	1.464	0.999
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	1.437	0.996	1.47	0.998	1.489	0.992	2.201	0.952	1.334	0.991
GaR-EEFR	1.327	0.988	1.375	0.984	1.418	0.969	2.087	0.923	1.231	0.969
GaR-RET	1.37	0.977	1.46	0.984	1.494	0.981	2.235	0.951	1.327	0.954
GaR-SMB	1.491	0.99	1.539	0.993	1.537	0.985	2.267	0.951	1.355	0.985
GaR-HML	1.585	0.989	1.65	0.996	1.633	0.997	2.433	0.966	1.433	0.996
GaR-MOM	1.664	0.97	1.698	0.975	1.729	0.972	2.453	0.959	1.511	0.948
GaR-VXO	1.202	0.935	1.251	0.961	1.3	0.96	1.918	0.927	1.107	0.892
GaR-CSPREAD	1.3	0.995	1.307	0.991	1.27	0.946	1.819	0.871	1.115	0.984
GaR-TERM	1.438	0.995	1.471	0.997	1.486	0.99	2.18	0.948	1.318	0.99
GaR-TED	1.407	0.994	1.454	0.993	1.465	0.984	2.113	0.935	1.276	0.987
GaR-ADS	1.089	0.796	0.788	0.247	0.762	0.248	0.764	0.231	0.978	0.446
GaR-CISS	1.169	0.973	1.22	0.995	1.243	0.978	1.792	0.906	1.06	0.799

Sources: FRED database and authors' computations.

Note: This table shows the tick loss for each individual GaR relative to the combined GaR for different daily horizons. Bold values represent the best model across the models considered for each daily horizon h_d. Diebold and Mariano t-test (DM) to test equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark.

Table B2. Out-of-sample results of BMIDAS-Q: Relative tick loss of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	2.133	0.972	2.109	0.971	2.075	0.97	2.054	0.97	2.026	0.967
GaR-EEFR	1.892	0.969	1.842	0.965	1.802	0.964	1.799	0.965	1.685	0.956
GaR-RET	1.259	0.966	1.348	0.989	1.331	0.975	1.333	0.969	1.346	0.973
GaR-SMB	2.217	0.978	2.215	0.979	2.16	0.981	2.181	0.977	2.152	0.977
GaR-HML	2.15	0.965	2.227	0.965	2.265	0.954	2.432	0.948	2.058	0.942
GaR-MOM	2.706	0.979	2.546	0.983	2.325	0.984	2.101	0.972	2.1	0.98
GaR-VXO	1.378	0.959	1.36	0.965	1.335	0.97	1.311	0.971	1.245	0.957
GaR-CSPREAD	1.561	1	1.544	1	1.528	1	1.519	1	1.493	1
GaR-TERM	2.163	0.973	2.128	0.971	2.089	0.97	2.068	0.97	2.056	0.969
GaR-TED	1.488	0.917	1.458	0.916	1.421	0.914	1.395	0.93	1.374	0.935
GaR-ADS	1.177	0.908	1.197	0.935	1.125	0.884	0.961	0.33	1.119	0.784
GaR-CISS	1.651	1	1.623	1	1.585	0.999	1.554	0.999	1.529	0.999
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	1.342	0.989	1.368	0.991	1.367	0.977	2.054	0.923	1.245	0.972
GaR-EEFR	1.232	0.982	1.248	0.983	1.26	0.96	1.909	0.91	1.135	0.934
GaR-RET	1.183	0.951	1.26	0.969	1.248	0.947	1.834	0.877	1.117	0.98
GaR-SMB	1.392	0.993	1.414	0.991	1.403	0.975	2.15	0.928	1.295	0.983
GaR-HML	1.435	0.983	1.491	0.99	1.496	0.985	2.268	0.945	1.304	0.975
GaR-MOM	1.51	0.991	1.488	0.995	1.445	0.994	2.094	0.933	1.285	0.99
GaR-VXO	1.039	0.72	1.067	0.894	1.112	0.927	1.704	0.892	1.009	0.545
GaR-CSPREAD	1.194	0.991	1.227	0.998	1.229	0.985	1.85	0.902	1.127	1
GaR-TERM	1.354	0.99	1.377	0.991	1.376	0.977	2.073	0.925	1.26	0.977
GaR-TED	1.242	0.969	1.263	0.969	1.257	0.935	1.885	0.882	1.142	0.964
GaR-ADS	1	0.501	0.754	0.204	0.742	0.215	0.689	0.196	0.907	0.305
GaR-CISS	1.139	0.979	1.177	1	1.191	0.995	1.791	0.913	1.073	0.859

Sources: FRED database and authors' computations.

Note: This table shows the tick loss for each individual GaR relative to the combined GaR for different daily horizons. Bold values represent the best model across the models considered for each daily horizon h_d. Diebold and Mariano t-test (DM) to test equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark.

Table B3. Out-of-sample results of EN-Q: Relative tick loss of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	1.339	0.988	1.327	0.992	1.305	0.991	1.462	0.981	1.456	0.976
GaR-EEFR	1.373	0.959	1.569	0.919	1.41	0.956	1.564	0.969	1.549	0.959
GaR-RET	1.589	0.992	1.536	0.971	1.919	0.992	1.732	0.98	2.263	0.986
GaR-SMB	1.308	1	1.362	1	1.611	0.992	1.756	0.997	1.681	0.996
GaR-HML	1.631	0.991	1.182	0.925	1.289	0.993	1.854	0.997	1.809	0.993
GaR-MOM	1.05	0.808	1.355	0.95	1.312	0.989	1.87	0.984	1.45	0.951
GaR-VXO	1.315	0.974	1.328	0.979	1.293	0.97	1.257	0.89	1.284	0.924
GaR-CSPREAD	1.17	0.907	1.177	0.934	1.144	0.911	1.303	0.961	1.25	0.955
GaR-TERM	1.273	0.987	1.255	0.993	1.253	0.994	1.427	0.974	1.398	0.971
GaR-TED	1.2	0.83	1.195	0.853	1.181	0.855	1.364	0.915	1.261	0.845
GaR-ADS	1.038	0.589	1.038	0.588	0.96	0.405	0.865	0.23	1.048	0.672
GaR-CISS	1.046	0.585	1.022	0.539	1.073	0.634	1.172	0.727	1.165	0.723
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	1.165	0.967	1.151	0.988	1.33	0.955	1.926	0.906	1.202	0.993
GaR-EEFR	1.216	0.981	1.261	0.977	1.382	0.953	1.983	0.91	1.159	0.976
GaR-RET	1.414	0.988	1.381	0.988	1.649	0.972	1.992	0.911	1.414	0.991
GaR-SMB	1.203	0.98	1.429	0.989	1.409	0.986	2.108	0.921	1.268	0.992
GaR-HML	1.367	0.994	1.436	0.98	1.179	0.975	2.634	0.951	1.266	0.996
GaR-MOM	1.222	0.952	1.374	0.982	1.439	0.949	2.247	0.931	1.196	0.973
GaR-VXO	1.078	0.926	1.127	0.992	1.264	0.931	1.839	0.877	1.135	0.959
GaR-CSPREAD	1.152	0.952	1.132	0.972	1.293	0.877	1.845	0.872	1.114	0.928
GaR-TERM	1.145	0.945	1.157	0.985	1.35	0.958	1.988	0.918	1.187	0.989
GaR-TED	1.14	0.911	1.155	0.96	1.316	0.903	1.927	0.885	1.132	0.923
GaR-ADS	0.961	0.38	0.809	0.169	0.672	0.146	0.733	0.19	0.845	0.316
GaR-CISS	1.05	0.757	1.032	0.67	1.216	0.847	1.775	0.861	1.091	0.827

Sources: FRED database and authors' computations.

Note: This table shows the tick loss for each individual GaR relative to the combined GaR for different daily horizons. Bold values represent the best model across the models considered for each daily horizon h_d. Diebold and Mariano t-test (DM) to test equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark.

Table B4. Out-of-sample results of LASSO-PCA-Q: Relative tick loss of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	2.2	0.978	2.26	0.988	1.868	0.988	1.723	0.997	1.678	0.996
GaR-EEFR	2.254	0.961	2.295	0.957	2.074	0.973	1.795	0.974	1.822	0.976
GaR-RET	2.222	0.98	2.27	0.968	1.777	0.927	2.197	0.984	1.544	0.978
GaR-SMB	2.288	0.992	1.851	0.971	1.826	0.988	1.649	0.981	1.588	0.989
GaR-HML	2.163	0.94	2.425	0.957	1.938	0.96	2.188	0.988	1.726	0.942
GaR-MOM	1.475	0.995	1.958	0.95	1.622	0.988	1.976	0.971	1.7	0.989
GaR-VXO	1.785	0.99	1.714	0.994	1.591	0.999	1.328	0.94	1.22	0.867
GaR-CSPREAD	2.271	0.982	1.939	0.996	1.406	1	1.241	0.915	1.227	0.931
GaR-TERM	2.204	0.986	2.105	0.987	1.816	0.987	1.73	0.992	1.656	0.99
GaR-TED	1.644	0.923	1.736	0.965	1.562	0.965	1.407	0.951	1.379	0.933
GaR-ADS	1.187	0.847	1.186	0.834	0.928	0.307	0.943	0.374	1.228	0.884
GaR-CISS	1.621	0.999	1.567	0.998	1.416	0.99	1.205	0.775	1.167	0.734
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	1.418	0.964	1.807	0.994	2.591	0.952	2.139	0.957	1.319	0.977
GaR-EEFR	1.502	0.969	1.938	0.988	2.834	0.958	2.284	0.955	1.445	0.975
GaR-RET	1.392	0.992	1.743	0.974	2.423	0.909	2.07	0.947	1.171	0.981
GaR-SMB	1.239	0.908	1.602	0.952	2.201	0.926	1.857	0.893	1.227	0.995
GaR-HML	1.314	0.971	2.051	0.99	2.658	0.964	3.143	0.963	1.209	0.968
GaR-MOM	1.14	0.882	1.99	0.983	2.444	0.912	2.059	0.926	1.208	0.992
GaR-VXO	1.087	0.7	1.48	0.983	2.122	0.921	1.688	0.881	1.09	0.963
GaR-CSPREAD	1.271	0.963	1.532	0.983	2.139	0.888	1.725	0.874	1.079	0.981
GaR-TERM	1.377	0.956	1.742	0.995	2.385	0.94	2.091	0.952	1.3	0.984
GaR-TED	1.274	0.932	1.662	0.988	2.45	0.934	2.018	0.93	1.247	0.978
GaR-ADS	1.499	0.851	0.606	0.139	0.61	0.113	0.605	0.122	0.902	0.301
GaR-CISS	1.063	0.737	1.365	0.959	2.05	0.884	1.685	0.867	1.079	0.88

Sources: FRED database and authors' computations.

Note: This table shows the tick loss for each individual GaR relative to the combined GaR for different daily horizons. Bold values represent the best model across the models considered for each daily horizon h_d. Diebold and Mariano t-test (DM) to test equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark.

Table B5. Out-of-sample results of EN-PCA-Q: Relative tick loss of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	2.232	0.978	2.072	0.99	1.768	0.988	1.772	0.997	1.692	0.996
GaR-EEFR	2	0.965	2.392	0.935	1.819	0.972	1.711	0.971	1.649	0.968
GaR-RET	2.107	0.973	2.049	0.962	1.792	0.946	2.2	0.994	1.594	0.971
GaR-SMB	2.079	0.991	2.202	0.991	1.833	0.994	1.93	0.986	2.069	0.99
GaR-HML	2.2	0.957	1.447	0.916	1.57	0.973	1.514	0.968	1.655	0.916
GaR-MOM	1.448	0.987	2.076	0.959	1.578	0.987	2.072	0.96	1.756	0.973
GaR-VXO	1.813	0.991	1.55	0.995	1.521	0.997	1.258	0.921	1.268	0.886
GaR-CSPREAD	2.242	0.979	1.789	0.993	1.367	1	1.275	0.963	1.259	0.951
GaR-TERM	2.198	0.984	1.946	0.991	1.728	0.989	1.754	0.99	1.684	0.989
GaR-TED	1.643	0.918	1.603	0.968	1.485	0.964	1.44	0.96	1.397	0.931
GaR-ADS	1.678	0.95	1.496	0.91	1.175	0.717	0.939	0.375	1.263	0.915
GaR-CISS	1.62	0.999	1.448	0.99	1.349	0.961	1.236	0.824	1.185	0.749
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	1.445	0.968	1.555	0.995	1.53	0.99	1.837	0.973	1.346	0.983
GaR-EEFR	1.449	0.982	1.707	0.992	1.55	0.987	1.855	0.963	1.367	0.979
GaR-RET	1.408	0.992	1.556	0.989	1.503	0.976	1.776	0.964	1.208	0.957
GaR-SMB	1.271	0.948	1.51	0.989	1.301	0.991	1.667	0.96	1.302	0.996
GaR-HML	1.453	0.995	1.504	0.983	1.281	0.913	1.834	0.942	1.3	0.975
GaR-MOM	1.274	0.981	1.712	0.971	1.51	0.969	1.722	0.934	1.307	0.988
GaR-VXO	1.196	0.908	1.335	0.995	1.317	0.994	1.426	0.893	1.129	0.95
GaR-CSPREAD	1.336	0.991	1.351	0.993	1.28	0.939	1.501	0.888	1.133	0.956
GaR-TERM	1.42	0.974	1.502	0.995	1.432	0.993	1.789	0.971	1.334	0.989
GaR-TED	1.315	0.96	1.433	0.994	1.42	0.985	1.731	0.95	1.279	0.98
GaR-ADS	1.375	0.923	0.595	0.174	0.655	0.159	0.504	0.152	0.743	0.26
GaR-CISS	1.098	0.898	1.177	0.987	1.188	0.936	1.446	0.884	1.108	0.877

Sources: FRED database and authors' computations.

Note: This table shows the tick loss for each individual GaR relative to the combined GaR for different daily horizons. Bold values represent the best model across the models considered for each daily horizon h_d. Diebold and Mariano t-test (DM) to test equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark.

Appendix C: UC and DQ tests for alternative models

Table C1. Out-of-sample results of MIDAS-Q: UC and DQ tests of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	0.565	0.039	0.565	0.042	0.565	0.032	0.565	0.034	0.565	0.032
GaR-EEFR	0.273	0.003	0.095	0.142	0.273	0.002	0.273	0.003	0.273	0.004
GaR-RET	0.427	0.452	0.226	0.713	0.046	0.281	0.046	0.027	0.046	0.015
GaR-SMB	0.926	0.2	0.717	0.509	0.717	0.553	0.926	0.191	0.926	0.153
GaR-HML	0.565	0.56	0.565	0.561	0.717	0.506	0.717	0.077	0.926	0.273
GaR-MOM	0.926	0.142	0.926	0.054	0.565	0.147	0.273	0.368	0.095	0.141
GaR-VXO	0.926	0.903	0.717	0.878	0.717	0.867	0.717	0.863	0.226	0.887
GaR-CSPREAD	0.717	0.088	0.926	0.09	0.926	0.102	0.273	0.925	0.095	0.826
GaR-TERM	0.565	0.056	0.565	0.06	0.273	0.002	0.273	0.002	0.273	0.002
GaR-TED	0.226	0.273	0.107	0.265	0.226	0.309	0.226	0.31	0.226	0.323
GaR-ADS	0.226	0.229	0.226	0.201	0.226	0.23	0.006	0	0.107	0.072
GaR-CISS	0.273	0.909	0.273	0.911	0.926	0.578	0.273	0.896	0.565	0.919
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	0.547	0.001	0.547	0.001	0.547	0	0.547	0	0.547	0
GaR-EEFR	0.86	0.001	0.786	0.004	0.86	0	0.86	0	0.86	0
GaR-RET	0.074	0.059	0.031	0.015	0.004	0.005	0.004	0.001	0.004	0.001
GaR-SMB	0.312	0.004	0.16	0.014	0.16	0.015	0.312	0.003	0.312	0.003
GaR-HML	0.547	0.008	0.547	0.007	0.16	0.007	0.16	0.001	0.312	0.005
GaR-MOM	0.312	0.005	0.312	0	0.547	0.004	0.86	0.008	0.786	0
GaR-VXO	0.547	0.675	0.16	0.763	0.16	0.685	0.16	0.656	0.031	0.584
GaR-CSPREAD	0.16	0.1	0.312	0.035	0.312	0.056	0.86	0.626	0.786	0.405
GaR-TERM	0.547	0.001	0.547	0.001	0.86	0	0.86	0	0.86	0
GaR-TED	0.031	0.242	0.012	0.028	0.031	0.033	0.031	0.041	0.031	0.032
GaR-ADS	0.16	0.146	0.16	0.088	0.074	0.015	0.004	0.002	0.031	0.008
GaR-CISS	0.86	0.835	0.86	0.74	0.312	0.091	0.86	0.594	0.547	0.862

Sources: FRED database and authors' computations.

Note: This table shows the following two backtesting tests for different individual GaR nowcasts: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), in which the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Bold values indicate which model passes this two tests with 10% probability.

Table C2. Out-of-sample results of BMIDAS-Q: UC and DQ tests of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	0.095	0.139	0.095	0.141	0.095	0.142	0.095	0.141	0.273	0.012
GaR-EEFR	0.273	0.006	0.273	0.006	0.095	0.082	0.095	0.109	0.095	0.074
GaR-RET	0.427	0.058	0.926	0.682	0.565	0.877	0.427	0.638	0.717	0.383
GaR-SMB	0.565	0	0.565	0	0.565	0	0.565	0	0.565	0
GaR-HML	0.565	0.029	0.565	0.022	0.273	0.228	0.273	0.005	0.273	0.139
GaR-MOM	0.273	0.012	0.273	0.013	0.273	0.013	0.273	0.012	0.095	0.125
GaR-VXO	0.565	0.907	0.565	0.908	0.565	0.909	0.926	0.907	0.717	0.71
GaR-CSPREAD	0.095	0.885	0.095	0.881	0.095	0.88	0.095	0.874	0.019	0.811
GaR-TERM	0.095	0.139	0.095	0.142	0.095	0.142	0.095	0.14	0.273	0.012
GaR-TED	0.717	0.731	0.717	0.713	0.717	0.695	0.717	0.686	0.717	0.696
GaR-ADS	0.565	0.892	0.926	0.901	0.565	0.758	0.095	0.432	0.095	0.375
GaR-CISS	0.565	0.704	0.565	0.708	0.273	0.856	0.273	0.863	0.565	0.725
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	0.455	0.015	0.455	0.015	0.455	0.015	0.455	0.014	0.786	0.004
GaR-EEFR	0.786	0	0.786	0	0.455	0.001	0.455	0.004	0.455	0.002
GaR-RET	0.16	0.14	0.547	0.163	0.86	0.362	0.16	0.168	0.312	0.052
GaR-SMB	0.86	0	0.86	0	0.86	0	0.86	0	0.86	0
GaR-HML	0.86	0.001	0.86	0	0.786	0.007	0.786	0.001	0.786	0.05
GaR-MOM	0.786	0.003	0.786	0.003	0.786	0.004	0.786	0.004	0.455	0.009
GaR-VXO	0.786	0.864	0.86	0.462	0.86	0.68	0.547	0.709	0.312	0.119
GaR-CSPREAD	0.455	0.875	0.455	0.86	0.455	0.856	0.455	0.856	0.208	0.616
GaR-TERM	0.455	0.015	0.455	0.015	0.455	0.015	0.455	0.015	0.786	0.004
GaR-TED	0.312	0.328	0.312	0.327	0.312	0.32	0.312	0.315	0.312	0.324
GaR-ADS	0.86	0.716	0.547	0.529	0.547	0.835	0.208	0.569	0.455	0.372
GaR-CISS	0.86	0.879	0.86	0.859	0.786	0.913	0.786	0.917	0.86	0.86

Sources: FRED database and authors' computations.

Note: This table shows the following two backtesting tests for different individual GaR nowcasts: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), in which the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Bold values indicate which model passes this two tests with 10% probability.

Table C3. Out-of-sample results of EN-Q: UC and DQ tests of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	0.717	0.488	0.717	0.464	0.717	0.488	0.926	0.216	0.717	0.481
GaR-EEFR	0.926	0.099	0.565	0.147	0.926	0.002	0.926	0.124	0.926	0.073
GaR-RET	0.717	0.007	0.107	0	0.427	0	0.717	0.002	0.226	0.01
GaR-SMB	0.565	0.071	0.565	0.064	0.926	0.131	0.717	0.012	0.926	0.021
GaR-HML	0.107	0	0.717	0.032	0.926	0.032	0.427	0.003	0.427	0.017
GaR-MOM	0.565	0.001	0.273	0.648	0.107	0	0.427	0.007	0.427	0.134
GaR-VXO	0.427	0.656	0.427	0.645	0.226	0.358	0.427	0.583	0.926	0.922
GaR-CSPREAD	0.717	0.856	0.717	0.879	0.717	0.855	0.717	0.856	0.565	0.9
GaR-TERM	0.926	0.244	0.565	0.779	0.565	0.783	0.565	0.783	0.565	0.78
GaR-TED	0.107	0.33	0.046	0.235	0.107	0.329	0.018	0.068	0.107	0.521
GaR-ADS	0.565	0.741	0.926	0.678	0.565	0.573	0.019	0.286	0.565	0.889
GaR-CISS	0.565	0.938	0.565	0.94	0.926	0.834	0.926	0.853	0.926	0.85
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	0.312	0.145	0.312	0.144	0.16	0.028	0.312	0.007	0.16	0.015
GaR-EEFR	0.547	0.085	0.86	0.313	0.547	0.001	0.547	0.079	0.547	0.094
GaR-RET	0.312	0	0.031	0	0.16	0	0.312	0.003	0.074	0.001
GaR-SMB	0.86	0.011	0.547	0	0.547	0.038	0.312	0.002	0.547	0.016
GaR-HML	0.031	0	0.16	0	0.547	0.069	0.074	0	0.16	0.011
GaR-MOM	0.86	0	0.86	0.002	0.012	0	0.16	0.001	0.16	0.08
GaR-VXO	0.312	0.498	0.16	0.111	0.074	0.056	0.16	0.113	0.547	0.458
GaR-CSPREAD	0.312	0.402	0.312	0.389	0.312	0.42	0.312	0.384	0.86	0.426
GaR-TERM	0.312	0.014	0.547	0.129	0.547	0.088	0.547	0.074	0.547	0.072
GaR-TED	0.031	0.039	0.012	0.02	0.031	0.04	0.004	0.003	0.031	0.082
GaR-ADS	0.786	0.921	0.547	0.551	0.547	0.022	0.208	0.432	0.547	0.776
GaR-CISS	0.86	0.893	0.86	0.891	0.547	0.809	0.547	0.801	0.547	0.78

Sources: FRED database and authors' computations.

Note: This table shows the following two backtesting tests for different individual GaR nowcasts: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), in which the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Bold values indicate which model passes this two tests with 10% probability.

Table C4. Out-of-sample results of LASSO-PCA-Q: UC and DQ tests of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	0.926	0.226	0.717	0.227	0.926	0.16	0.926	0.069	0.565	0.05
GaR-EEFR	0.565	0.394	0.926	0.11	0.926	0.025	0.926	0.022	0.717	0.029
GaR-RET	0.107	0.05	0.717	0.364	0.926	0.905	0.006	0	0.018	0.007
GaR-SMB	0.427	0.154	0.717	0.239	0.926	0.279	0.717	0.435	0.565	0.089
GaR-HML	0.107	0.123	0.427	0.009	0.427	0.133	0.226	0.008	0.427	0.336
GaR-MOM	0.226	0.084	0.926	0.884	0.427	0.626	0.926	0.096	0.046	0.006
GaR-VXO	0.107	0.481	0.226	0.57	0.226	0.382	0.107	0.189	0.717	0.711
GaR-CSPREAD	0.717	0.146	0.717	0.148	0.926	0.923	0.926	0.925	0.565	0.929
GaR-TERM	0.273	0.003	0.273	0.003	0.926	0.2	0.565	0.079	0.273	0.678
GaR-TED	0.427	0.469	0.226	0.18	0.226	0.18	0.107	0.185	0.107	0.45
GaR-ADS	0.226	0.043	0.427	0.1	0.226	0.085	0.226	0.014	0.226	0.229
GaR-CISS	0.717	0.885	0.926	0.884	0.926	0.881	0.565	0.653	0.926	0.888
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	0.312	0.004	0.16	0.004	0.312	0.001	0.312	0.001	0.547	0.001
GaR-EEFR	0.547	0.009	0.312	0.002	0.312	0.001	0.312	0	0.16	0.002
GaR-RET	0.031	0.001	0.312	0.046	0.312	0.035	0	0	0.004	0.007
GaR-SMB	0.16	0.009	0.16	0.007	0.312	0.028	0.16	0.028	0.547	0
GaR-HML	0.031	0.082	0.074	0	0.074	0.005	0.031	0.003	0.074	0.041
GaR-MOM	0.031	0.038	0.312	0.031	0.074	0.136	0.547	0.037	0.004	0
GaR-VXO	0.031	0.066	0.074	0.163	0.074	0.074	0.031	0.023	0.312	0.372
GaR-CSPREAD	0.312	0.436	0.312	0.566	0.547	0.925	0.547	0.928	0.86	0.954
GaR-TERM	0.86	0	0.86	0	0.312	0.006	0.547	0.001	0.86	0.014
GaR-TED	0.074	0.018	0.031	0.002	0.031	0.001	0.012	0.001	0.012	0.015
GaR-ADS	0.074	0.001	0.312	0.095	0.074	0.023	0.16	0.021	0.031	0.004
GaR-CISS	0.312	0.785	0.547	0.787	0.547	0.81	0.86	0.77	0.547	0.814

Sources: FRED database and authors' computations.

Note: This table shows the following two backtesting tests for different individual GaR nowcasts: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), in which the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Bold values indicate which model passes this two tests with 10% probability.

Table C5. Out-of-sample results of EN-PCA-Q: UC and DQ tests of individual GaR nowcasts

	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Before COVID-19 (2007Q1 to 2019Q4)										
GaR- ISPREAD	0.717	0.422	0.717	0.256	0.926	0.159	0.926	0.061	0.565	0.054
GaR-EEFR	0.565	0.769	0.427	0.06	0.926	0.114	0.427	0.455	0.427	0.589
GaR-RET	0.427	0.042	0.427	0.507	0.226	0.061	0.006	0	0.006	0.002
GaR-SMB	0.427	0.607	0.046	0.001	0.226	0.035	0.226	0.219	0.926	0.009
GaR-HML	0.107	0.115	0.717	0.333	0.926	0.253	0.427	0.12	0.717	0.155
GaR-MOM	0.226	0.083	0.717	0.092	0.226	0.716	0.427	0.003	0.226	0.021
GaR-VXO	0.226	0.614	0.107	0.188	0.226	0.383	0.107	0.337	0.427	0.725
GaR-CSPREAD	0.717	0.147	0.717	0.148	0.926	0.919	0.926	0.925	0.565	0.93
GaR-TERM	0.565	0.062	0.273	0.003	0.926	0.198	0.565	0.074	0.273	0.68
GaR-TED	0.427	0.469	0.226	0.18	0.226	0.18	0.107	0.185	0.107	0.451
GaR-ADS	0.107	0.031	0.226	0.096	0.226	0.006	0.226	0.008	0.226	0.39
GaR-CISS	0.717	0.884	0.926	0.883	0.926	0.88	0.565	0.648	0.565	0.689
Including COVID-19 (2007Q1 to 2020Q4)										
GaR- ISPREAD	0.16	0.01	0.16	0.005	0.312	0.001	0.312	0.001	0.547	0.001
GaR-EEFR	0.547	0.026	0.074	0.002	0.312	0.003	0.074	0.011	0.074	0.018
GaR-RET	0.16	0.005	0.074	0.002	0.031	0	0	0	0.001	0
GaR-SMB	0.16	0.26	0.004	0	0.031	0	0.031	0.009	0.312	0
GaR-HML	0.012	0.001	0.16	0.003	0.312	0.049	0.074	0.014	0.16	0.012
GaR-MOM	0.031	0.254	0.16	0.003	0.031	0.032	0.16	0.009	0.031	0
GaR-VXO	0.031	0.074	0.012	0.004	0.031	0.02	0.031	0.053	0.16	0.388
GaR-CSPREAD	0.16	0.053	0.16	0.107	0.312	0.407	0.547	0.923	0.86	0.953
GaR-TERM	0.547	0.001	0.86	0	0.312	0.003	0.547	0.001	0.86	0.014
GaR-TED	0.074	0.018	0.031	0.002	0.031	0.001	0.012	0.001	0.012	0.016
GaR-ADS	0.074	0.029	0.074	0.006	0.074	0.001	0.074	0.003	0.031	0.109
GaR-CISS	0.312	0.786	0.547	0.787	0.547	0.811	0.86	0.767	0.86	0.688

Sources: FRED database and authors' computations.

Note: This table shows the following two backtesting tests for different individual GaR nowcasts: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), in which the null hypothesis is that the exceedance indicator (Hit) is an i.i.d. process (not rejecting is better). Bold values indicate which model passes this two tests with 10% probability.